

TÜRKİYE CUMHURİYETİ

ANKARA ÜNİVERSİTESİ

SOSYAL BİLİMLER ENSTİTÜSÜ

KAMU HUKUKU ANABİLİMDALI

LEGAL REALISM'S RENAISSANCE :

LEGAL DECISION MAKING WITH ARTIFICIAL INTELLIGENCE

YÜKSEK LİSANS TEZİ

EGEMEN BAHADIR DUMLUDAĞ

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Sosyal Bilimler Enstitüsü Müdürlüğüne

Prof. Dr. Gülriz UYGUR danışmanlığında hazırladığım “Legal Realism’s Renaissance: Legal Decision Making with AI (Ankara, 2024)” adlı yüksek lisans tezindeki bütün bilgilerin akademik kurallara ve etik davranış ilkelerine uygun olarak toplanıp sunulduğunu, başka kaynaklardan aldığım bilgileri metinde ve kaynakçada eksiksiz olarak gösterdiğimi, çalışma sürecinde bilimsel araştırma ve etik kurallarına uygun olarak davrandığımı ve aksinin ortaya çıkması durumunda her türlü yasal sonucu kabul edeceğimi beyan ederim.

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TABLE OF CONTENTS

TABLE OF CONTENTS.....	ii
ABBREVIATIONS.....	iv
INTRODUCTION	1
CHAPTER 1: THE LEGAL REALIST APPROACH TO LAW	6
I. LEGAL REALISM IN GENERAL.....	7
A. AN OVERVIEW	9
B. KARL NICKERSON LLEWELLYN.....	15
C. JEROME FRANK.....	18
II. NEW LEGAL REALISM.....	20
III. LEGAL REALISM IN PARTICULAR.....	27
A. WHAT LAW REFERS TO	29
B. WHAT LEGAL REALISM ENTAILS.....	36
CHAPTER 2: THE SILICON ORACLES	44
I. COURTROOM PREDICTIONS TODAY	46
A. DECISION TREES	47
B. SUPPORT VECTOR MACHINES	50
C. DEEP LEARNING.....	52
II. PATTERN RECOGNITION AND PROBABILITIES.....	54
CHAPTER 3: AI FROM THE LENS OF LEGAL REALISM	58
I. FIRST RESPONDERS.....	58

A. LEGAL DETERMINISM	58
B. A CLOSER INSPECTION	69
II. A COMPATIBLE COUPLE	74
A. WHERE AI MODELS LACK.....	76
B. APPLYING REALISM TO AI.....	79
CONCLUSION.....	86
REFERENCES	93
ABSTRACT	103
ÖZET	104

ABBREVIATIONS

AI: Artificial intelligence

DT: Decision Tree

SVM: Support Vector Machine

DL: Deep Learning

RNN: Recurrent Neural Network

CNN: Convoluted Neural Network

ECtHR: European Court of Human Rights

INTRODUCTION

There are discussions as to what law is. There are discussions as to what artificial intelligence is. However, both questions are dependent on how a person would like to define these concepts. It is entirely possible to define the word law as “a piece of furniture consisting of a smooth flat slab fixed on legs”¹. Then, we must ask, what would this definition achieve? The problem of definition is often followed in the wrong direction, we try and define words as we accept them in a pre-existing state, when in fact what we aim to do (or ought to aim to do) is to name concepts as they exist in the world around. As **Jerome Frank** puts it: “Definitions are only good if they are helpful.”²

From such a perspective, we need to highlight phenomena as they occur in the world. What is happening in the world is that there are studies that have managed to predict legal decisions with varying degrees of success using “artificial intelligence”. In order to make sense of this phenomenon, akin to all the other matters of the world, we need to ask questions about it. There are infinitely many questions that can be raised about this issue. In this thesis, we will limit our scope in trying to find an answer as to if and how American Legal Realism provides a basis for a legal theoretical framework regarding legal decision making with artificial intelligence.

In pursuit of the above question, we argue that Legal Realism has a perspective that is suitable for understanding the AI models used in legal decision making. Along with that, we also argue, as a sequitur of the main argument, if these AI models see widespread use, Legal Realism will have more ground in legal theory literature, what we call a renaissance of Legal Realism.

¹ Merriam-Webster, “Table”, 2023.

² Jerome Frank, “Are Judges Humans? Part One: The Effect on Legal Thinking of the Assumption That Judges Behave Like Human Beings,” *University of Pennsylvania Law Review*, Vol. 80, 1931, p. 45.

To that end, we have to answer two main questions: (1) “what is Legal Realism”, and (2) “what is artificial intelligence”. However, in the spirit we have mentioned above, these questions will seek not to define “Legal Realism” and “artificial intelligence” *per se*, but rather will aim to *describe* what Legal Realism and artificial intelligence entails for the scope of this thesis. As such, these questions are rephrased as “what do we mean by Legal Realism” and “what are the AI models used in legal decision prediction”.

In the structure of this work, the first chapter, “The Legal Realist Approach to Law”, focuses on the question “what do we mean by Legal Realism”. In this chapter, we first refer to what is understood by Realism in general by examining both primary and secondary sources and the debates around what Legal Realism is. After that, we investigate New Legal Realism and how it relates to Legal Realism. Finally, we try and pinpoint what we refer to when we talk of Legal Realist thought, as opposed to what it means in general. Legal Realism is subject to much debate, and we try and highlight certain aspects of it considering its main points and our subject.

The second chapter focuses on the current studies regarding legal decision prediction using AI. This chapter does not serve as an extensive survey of literature but rather as a catalogue of available technologies subject to study and how they perform. Using that catalogue, we then try to break down how those technologies operate at surface level as what matters to the scope of this thesis is what they are conceptually, not the way their algorithms are optimized. In other words, we try and understand what data they use and how they use it in their predictions.

The last chapter focuses on if and how Legal Realism is applicable to the operations of AI in legal decision prediction with references to previous chapters and other relevant text. First, we discuss pre-existing ideas from different authors pointing to the relationship between Legal Realism and AI in legal decision making. Then, we ask

more detailed questions as to how AI is compatible with the previously established method of study of the Legal Realist view, and how AI lacks in certain issues.

There is a significant lack of theoretical framework in legal studies regarding the analysis of AI usage in legal practice as decision making agents. There are many authors who have delved into the issue and concluded with certain normative approaches, saying that AI could be used as a tool of assistance, but it should not be used as an agent. Along with that, most studies focus on legality and ethics of AI. We believe, this will inevitably result in a crisis as AI applications become more common in practice, considering “the conception of society in flux, and in flux typically faster than the law”³.

The connection between Legal Realism and legal decision prediction by AI has caught attention of some authors but no definitive examination or exhaustive comparative study has been yet made. The relationship between AI and Legal Realism is mentioned in some AI papers, but Legal Realism is too large to investigate as a topic in scope, if the relationship itself is not made the main question of the research. Legal Realism and formalism has been discussed in an AI setting before but our approach differs as it is focused on application of Realist theory to AI in legal decision making.

Why is it so that, if Realism provides a framework for the analysis of legal decision prediction by these machines, or AI to be concise, we call this a renaissance? **Grant Gilmore** in 1961, writing of Realism points out that Realism was already a thing of the past in saying “realism was the academic formulation...”⁴, or **Theodore W. Ruger**

³ Karl N. Llewellyn, “Some Realism about Realism: Responding to Dean Pound”, *Harvard Law Review*, vol. 44, no. 7, June 1931, p. 1236.

⁴ Grant Gilmore, “Legal Realism: Its Cause and Cure”, *The Yale Law Journal*, Vol. 70, No. 7, June 1961, p. 1037.

and others mention that Realism has waned in the law schools.⁵ Legal Realism is not as popular as other legal theories, and there are *some* numbers to support that claim. A comparison between Legal Realism and legal positivism through Google Ngram Viewer yields the following Figure 1⁶, and comparison of their search popularity through Google Trends yields the following Figure 2⁷. However, considering the current interest in artificial intelligence as a research subject, any theory that provides a legal framework for the new technologies will have increased standing.

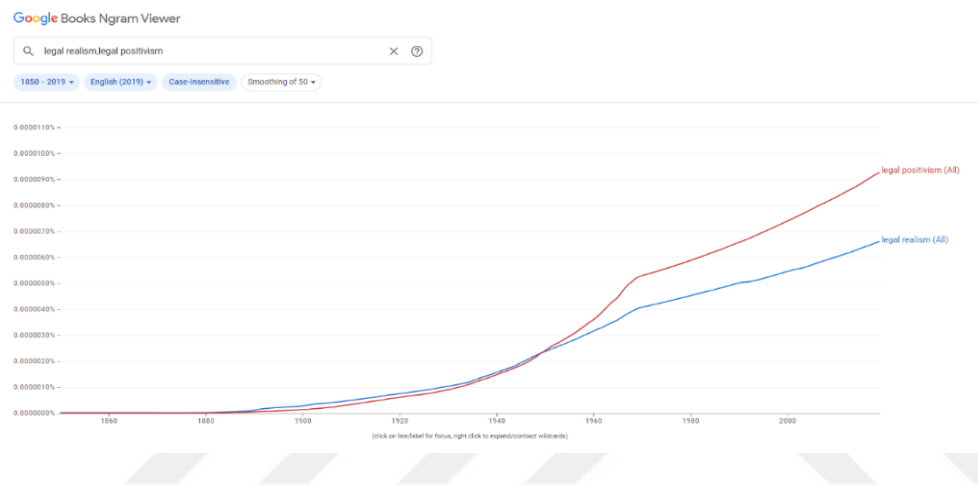


Figure 1: Google Ngram Viewer

⁵ Theodore W. Ruger et al., “The Supreme Court Forecasting Project: Legal and Political Science Approaches to Predicting Supreme Court Decision making”, *Columbia Law Review*, 2004, 104, p. 1157.

⁶ <http://books.google.com/ngrams>, search terms: legal realism, legal positivism, case-insensitive search.

⁷ <https://www.google.com/trends>, search terms: Legal realism, Legal positivism, between dates 01.01.2005-01.01.2023.

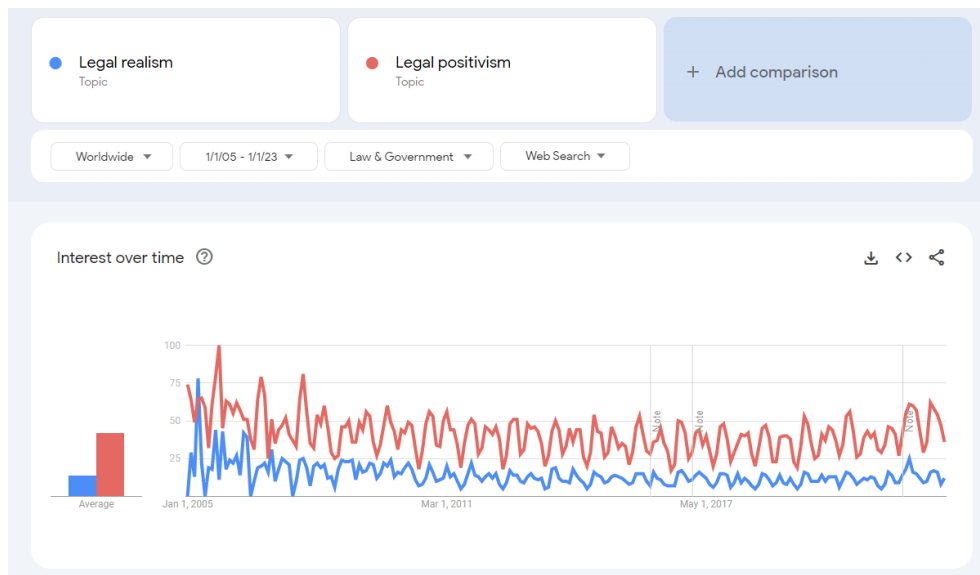


Figure 2: Google Trends between 01.01.2005-01.01.2023

The scope of this thesis does not include a general overview and criticism of either Legal Realism or usage of AI in legal decision making. What we aim to provide is a simple comparison showing how Legal Realism is compatible with the current models. As such, the sections dealing with these issues will be limited in scope. We aim to avoid normative comments and try to focus on how these phenomena are a compatible couple and how the success of AI practices in legal decision making may also bring success to Legal Realism as a theory.

CHAPTER 1:

THE LEGAL REALIST APPROACH TO LAW

We try and explore the Realist approach to law in three main sections in this chapter. In arguing that Legal Realism provides a suitable framework for understanding legal decision making with AI, we need to first make it clear what we mean by Legal Realism. There is more than one way to understand Realism, and we have to clarify what we understand by Legal Realism, and what we seek to convey in speaking of it. In pursuit of this the first section explores the concept of Legal Realism, the prominent writers and where they had stood at the height of the Realist debate. We focus on **Frank** and **Karl N. Llewellyn** as the prominent figures of Realism.

The next section discusses how New Legal Realism⁸ relates to Legal Realism. New Legal Realism itself might be considered a rebirth of Legal Realism. However, it has to be noted even New Legal Realism does not have significant traction in legal study discussions, and arguably even less so than Legal Realism itself, though this is not for us to discuss. In this section we thus discuss New Legal Realism as it relates to Legal Realism.

The last section pinpoints what we aim to mean by Legal Realism. While the first section summarises the main points of Legal Realist thought, Legal Realism is not monolithic and covers ideas in a wide range of topics. As such, there is a need to understand which ideas are important for the scope of the thesis, and the reasons as to why we consider these important. Along with that, this section also discusses what legal

⁸ In most of the publications we have referenced in this work, words “New Legal Realism” is often capitalised, while the practice as to “Legal Realism” is not as coherent. As such, we have decided to capitalise Legal Realism (or any referral to it such as Realism or Realists) in order to be consistent with the mentions of New Legal Realism.

decision making means in terms of Realism as the subject of the thesis is “legal decision making” with AI.

As the section will continue on to discuss, Realism is not a uniform school of thought, and the Realists were lawyers who sought an understanding of law that reflected what happened in the legal practice more accurately. There are countless Legal Realists in the literature⁹, yet taking all of them into consideration would be impossible and unnecessary for the purposes of this study. There are countless works serving as dedicated discussions of Legal Realism. **Llewellyn** has already created a survey of Legal Realists in his response to **Dean Pound**, therefore we believe a general survey of literature is made redundant.¹⁰ As the question of the thesis relates only to the fundamental ideas of Realism, we believe further works may be considered out of scope.

I. LEGAL REALISM IN GENERAL

*Cocoon above! Cocoon below!
Stealthy Cocoon, why hide you so
What all the world suspect?
An hour, and gay on every tree
Your secret, perched in ecstasy
Defies imprisonment!*¹¹

Llewellyn, in his reply to **Dean Pound**, notes that “There is no school of realists”¹². **Mark Tebbit** reiterates this by saying “There was no single outlook shared by all the Realists, and there was no conscious attempt to formulate a technically precise

⁹ 20, by Llewellyn’s final account in his reply.

¹⁰ Although, the list in this reply is subject to debate see. *Roscoe Pound and Karl Llewellyn: Searching for an American Jurisprudence* (1997) by Hull.

¹¹ First stanza of the poem “Cocoon above! Cocoon below!” by Emily Dickinson, depicting the metamorphosis of a butterfly. Emily Dickinson, “Cocoon above! Cocoon below!”, No. 129, 1935, *The Complete Poems of Emily Dickinson*, Ed. Thomas H. Johnson, Boston: Little, Brown & Company, 1960, p. 60.

¹² Llewellyn, 1931, p. 1233.

concept of law.”¹³ This view that there is no Realist school is also shared by **Frank**.¹⁴ This is an issue that makes it particularly difficult to create a list of markers of Legal Realist thought for any study.

Ülker Gürkan, in her survey of Legal Realism, notes **Frank** and **Llewellyn** as the leaders of particular understandings in Legal Realism, which she names as extremists and moderates respectively, in regard to their approach with behaviourism.¹⁵ The idea is echoed in **Tebbit** as he names **Frank** and **Llewellyn** as the “two leading lights” of “realism proper”.¹⁶ Along with that, **Llewellyn** (and **Frank**¹⁷) also has surveyed the Legal Realist ferment, and put forward a “summary” of the Legal Realists of the time. **Elizabeth Mertz** quotes:

In 1930, two giants in the US legal tradition heralded the onset of a new approach to studying law. Karl Llewellyn published an article listing twenty scholars as “legal realists”, while Jerome Frank referred to “legal realism” in his signal volume *Law and the Modern Mind*. Along with other legal theorists, Brian Tamanaha has convincingly argued that the loose collection of ideas surrounding the debates of that time had some shared emphases that cohere to form an important strand of jurisprudence.¹⁸

¹³ Mark Tebbit, *Philosophy of Law: An Introduction*, Third Edition, Routledge, 2017, p. 61.

¹⁴ Ülker Gürkan, *Hukuki Realizm Akımı*, Ankara: Sevinç Matbaası, 1967, p. 47.

¹⁵ Gürkan, p. 3.

¹⁶ Tebbit, p. 62.

¹⁷ Frank did not give his permission for his name to be signed as to give the credit to Llewellyn (LLEWELLYN, 1931, p. 1222n1).

¹⁸ Elizabeth Mertz, “Realism then and now: Using the real world to inform formal law”, *Research Handbook on Modern Legal Realism*, ed. Shauhin Talesh, Elizabeth Mertz, Heinz Klug, Elgaronline, 2021, p. 21.

On the other hand, **Brian Z. Tamanaha** argues “individuals identified as legal realists should not be seen as a group whose views define realism.”¹⁹ It is true that Realism might refer to a general approach; however, we consider Legal Realism as it refers to the views of Legal Realists in the spirit we have put forth in the beginning, we have to *describe* instead of *define* in order to engage in meaningful debate.

In considering how **Llewellyn** and **Frank** are considered to be the most prominent Legal Realists of their time, and how they have also surveyed the surrounding Legal Realist thought, the scope of our discussion will revolve mostly around the ideas of these two, and what they present as the answer to the question: “what is legal realism”. This is done so because it is both redundant to resurvey the Legal Realist thought, as it has already been surveyed by **Llewellyn** and many others, and because the discussion of Legal Realism in detail must be made a question of its own. This section will try and first give a general overview of what Legal Realism has come to refer to, broadly, and then try and summarize the writings of **Frank**, and **Llewellyn**.

A. AN OVERVIEW

There are a great many ways to categorise Legal Realist thought. We can study Midwest Legal Realism as opposed to East Coast Legal Realism, first wave and second wave Legal Realism, traditional Legal Realism and scientific Legal Realism, rule-sceptics and fact-sceptics, and so forth. However, as we discuss in the following section, Legal Realists have common ground, and Legal Realism means something.

¹⁹ Brian Z. Tamanaha, “Legal Realism in Context”, *The New Legal Realism, Volume I: Translating Law-and-Society for Today’s Legal Practice*, ed. Elizabeth Mertz, Stewart Macaulay, Thomas W. Mitchell, New York: Cambridge University Press, 2016, p. 164.

What does Legal Realism mean, then? For **Emily Taylor Poppe**, “Legal Realism was a multi-dimensional intellectual movement, encompassing a legal philosophy, a scholarly framework, and an approach to legal education, as well as ties to progressive social reforms.”²⁰ For **Laure Kalman** “The realists pointed to the indeterminacy of law. They distinguished between the judge’s decision resolving a legal dispute and his rationalization of the result in the opinion”.²¹ **Kalman** also notes that “The realists had put interdisciplinary legal research and education front and square on the agenda at the same time that they ensured social science took a back seat to law.”²²

Hanoch Dagan highlights the critical nature of Legal Realism in saying that it “beings with a critique of this formalist conception of law.”²³ **Dagan** then points to the point of the criticism, saying that they argued “formalist enterprise” fails to describe the adjudicative process.²⁴ **Kalman** also notes this point of critical nature by saying that “The realists made legal scholarship fun — if you were not one of their targets”²⁵.

Brian Leiter notes core claim of Realism— that is “judges respond primarily to the stimulus of facts.”²⁶²⁷²⁸ **Leiter** also uses this core claim to distinguish Realists from

²⁰ Emily Taylor Poppe, “New Legal Realism goes to law school: Integrating social science and law through legal education”, *Research Handbook on Modern Legal Realism*, ed. Shauhin Talesh, Elizabeth Mertz, Heinz Klug, Elgaronline, 2021, p. 191.

²¹ Laure Kalman, “East Coast Legal Realism and its progeny”, *Research Handbook on Modern Legal Realism*, ed. Shauhin Talesh, Elizabeth Mertz, Heinz Klug, Elgaronline, 2021, p. 39.

²² Kalman, p. 43.

²³ Hanoch Dagan, “The Realist Conception of Law”, *The University of Toronto Law Journal*, Vol. 57, No. 3, Summer 2007, p. 612.

²⁴ Dagan, p. 612.

²⁵ Kalman, p. 39.

²⁶ Brian Leiter, “Rethinking Legal Realism: Toward a Naturalized Jurisprudence”, *Texas Law Review*, Vol. 76, Issue 2, December 1997, p. 275.

²⁷ Brian Leiter, “Legal Realism and Legal Positivism Reconsidered”, *Ethics*, Vol. 111, No. 2, Jan. 2001, p. 281.

²⁸ Brian Leiter, “Legal Realism and Legal Doctrine”, *University of Pennsylvania Law Review*, Vol. 163, 2015, p. 1980.

the formalists in branding formalists descriptive claim as judges being primarily rule-responsive, while for Realists the judges are primarily fact-responsive.²⁹ He also distinguishes Realism as it is and what he calls the received view of Legal Realism:

According to what I will call the "Received View," Legal Realism is fundamentally: (1) a descriptive theory about the nature of judicial decision, according to which, (2) judges exercise unfettered discretion, in order (3) to reach results based on their personal tastes and values, which (4) they then rationalize after-the-fact with appropriate legal rules and reasons.³⁰

The "received view" is important to note. It represents what Realism is thought to be by many. **Leiter** argues that the scepticism in the received view is but one interpretation of the core claim he proposes.³¹ For **Leiter** Realists do not claim that judges exercise unfettered discretion but argued that judicial decisions "fall into discernible patterns (making prediction possible), though the patterns are not those one would expect from the existing legal rules."³²

Kalman reiterates the concern found in **Leiter**, about the branding of Realism as a nihilist approach:

Traditionalists misleadingly portrayed the realists as nihilists who undermined the usefulness of legal rules without putting anything in their place. So, too, orthodox scholars said the realists rendered judicial opinions incoherent, explained them on

²⁹ Leiter, 1997, p. 269.

³⁰ Leiter, 1997, p. 268.

³¹ Leiter, 1997, p. 268.

³² Leiter, "Legal Realism and Legal Positivism Reconsidered", 2001, p. 281.

the basis of what judges ate for breakfast, and imbued all decisions of authorities, no matter how illegitimate, with the authority of law.³³

Leiter also notes Realists argued that the “the judicial response to the “situation type”—that is, the distinctive factual pattern—that determines the outcome of the case.”, driving the point Realists argued for something more than simple nihilism forward.³⁴ On the other hand **Omar Zaky** notes “When analyzing the realist claim, it is important not to view their position as simply emphasizing the role of case facts to the decision, but as an empirical attempt at making the decision making process more transparent.”³⁵

To talk about Realism in a grounded fashion, we also need to inquire its roots. We have noted the view that there is no Realist school. However, we have also noted that there is a legalist approach which is united under some common points. **Leiter** quotes:

It is a commonplace—so oft repeated that it now has the status of dogma—that Legal Realism cannot be defined, that the movement is too disparate in its concerns to be characterized coherently. If one actually reads the Realists, however—as opposed to reading about them— the opposite turns out to be the case. For those writers who are, by any account, major figures in Legal Realism—for example, Karl Llewellyn, Jerome Frank, Max Radin, Underhill Moore, Hessel Yntema, Felix Cohen, Herman Oliphant, Leon Green, Joseph Hutcheson—all shared an interest in understanding judicial decision making and, in particular, shared certain substantive views about how adjudication really works.³⁶

³³ Kalman, p. 40.

³⁴ Leiter, “Legal Realism and Legal Positivism Reconsidered”, 2001, p. 281.

³⁵ Omar Zaky, *Finding Neverland? Artificial Intelligence and the Jurisprudence of Legal Certainty*, American University in Cairo Master’s Thesis, 2021, p. 45.

³⁶ Leiter, “Legal Realism and Legal Positivism Reconsidered”, 2001, p. 280.

This is what **Llewellyn** pointed out as well, Realists are united in certain areas even if they are not a school and seem too diverse.³⁷ There is a Legal Realism, and we need to actually read the Realists to see it.

Tebbit names **Oliver W. Holmes** and **John Chipman Gray** as the initiators of Legal Realism, and notes some name them proto-Realists, while others consider them early Realists.³⁸ **Gürkan**, considers **Holmes** to have inspired Realists with his scepticism.³⁹ Realists were not actually a school but as **Llewellyn** would call them, a ferment.⁴⁰⁴¹ Realists have turned their eyes towards the criticism of old theories and perspective and to the law in progress instead of the law in books.⁴²

One point of interest for rRealists was what the judges did do, in fact.⁴³ For **Gray** this took the form of a proposition as to what law is, for **Gray** law was what the judges did do and laws and precedents and all principles and rules were only the sources of law,⁴⁴ which is why this work is going to exclude **Gray** in Realist analysis, to try and manoeuvre around the debates about what law is. **Joseph Chappell Hutcheson Jr.**, has noted while the formalists argue that judges reach decisions with syllogism using laws, they actually make decisions first and then rationalise those decisions.⁴⁵ **Frank** has continued this attack on mechanical approach and formalism and argued that legal decision making does

³⁷ Llewellyn, 1931, p. 1223.

³⁸ Tebbit, p. 61-62.

³⁹ Gürkan, p. 27.

⁴⁰ Gürkan, p. 27

⁴¹ Llewellyn, 1931, p. 1222.

⁴² Gürkan, p. 28

⁴³ Tebbit, p. 63.

⁴⁴ Tebbit, p. 63.

⁴⁵ Tebbit, p. 69.

not differ from other decision making processes, **Frank** called this the “dominance of the conclusion”.⁴⁶⁴⁷

Another point of interest of Realist literature is scepticism towards laws, many Realists have defended that while most decisions were unpredictable if not arbitrary.⁴⁸ **Llewellyn** says that the problem is how judiciary differs from pre-existing rules.⁴⁹ **Frank** argues that rules are only informative, not constitutive.⁵⁰ **Llewellyn** shares this view but notes that in some cases the courts following these rules might be what “ought to” happen.⁵¹ **Llewellyn** says that paper rules act as normative prescriptions for the officials and that they tell the officials what the officials must do, and even in a behavioural analysis their existence cannot be denied.⁵²

Some authors consider **Llewellyn**’s “About Realism—Responding to Dean Pound” to be a manifesto of Realism,⁵³⁵⁴ although **Harry W. Jones** notes **Llewellyn** could consider “A realist Jurisprudence—The Next Step” a real manifesto.⁵⁵ In the first article, **Llewellyn** puts forward the common ground of the Realists of the period as they differed from their contemporaries.⁵⁶ He emphasizes some of these as characteristic to the movement (the fourth, the sixth, the seventh, the eighth, and the ninth)⁵⁷; which

⁴⁶ Tebbit, p. 70.

⁴⁷ Jerome Frank, *Law and the Modern Mind*, London: Stevens & Sons Limited, 1949, p. 101.

⁴⁸ Tebbit, p. 73.

⁴⁹ Karl N. Llewellyn, “A Realistic Jurisprudence — The Next Step”, *Columbia Law Review*, vol. 30, no. 4, Nis. 1930, p. 444.

⁵⁰ Tebbit, p 74.

⁵¹ Llewellyn, 1930, p. 448.

⁵² Llewellyn, 1930, p. 449.

⁵³ Tebbit, p. 73.

⁵⁴ Harry W. Jones, “Law and Morality in the Perspective of Legal Realism”, *Columbia Law Review*, Vol. 61, No. 5, May 1961, p. 799.

⁵⁵ Jones, p. 799n1.

⁵⁶ Llewellyn, 1931, p. 1235.

⁵⁷ Llewellyn, 1931, p. 1238.

essentially refer to a temporary split between “is” and “ought”, distrust towards the claim that traditional rules express what people do in fact, distrust towards the claim that rules are the fundamental basis of action in court decisions, the belief that cases and legal situations should be grouped more narrowly than before, the necessity that law should be examined in its impact, and the pragmatist attack in these lines towards the issues in law.⁵⁸ Some of these are also present in “A realist Jurisprudence—The Next Step” such as the necessity for more narrow categories.⁵⁹

This list by **Llewellyn** had been prepared in the early days of Legal Realism, **Gürkan** argues that the common ground of Realists, considering the later works, can be summarised as such: (1) distrust towards current legal perspectives, concepts, and rules, (2) the analysis of law in work, and explanation of law by the behaviour of courts, (3) an effort to establish scientific method in law, and (4) resort to psychology and psychological perspectives.⁶⁰

This overview provides us with some key points to keep in mind when talking about legal Realism. It seems possible to boil Realism down to the four points **Gürkan** has put forward. However, summarising **Llewellyn** and **Frank** is a must as they are the leading figures of the Realist view. In the following sections, we will cover **Llewellyn** and **Frank**, starting with the former and try and provide a more detailed summary of their ideas, albeit we aspire no more than to provide but a summary in this section.

B. KARL NICKERSON LLEWELLYN

⁵⁸ Llewellyn, 1931, p. 1236-1238.

⁵⁹ Llewellyn, 1930, p. 457.

⁶⁰ Gürkan, p. 31-32.

Realism, according to **Llewellyn**, has the motivation to establish an effective and objective technology for understanding and creating of the law.⁶¹ This would refer to the pragmatist spirit in the Realist vision of law. His idea is that Realists differ in their interests yet are united in their angles of attack.⁶²

Llewellyn lays out the program of Realist work, including areas such as: interdisciplinary work with psychology, examination of rules, recategorization of facts, study of relationships between facts and legal material as they impact the thinking of the judge, study of legal decision prediction, and so forth.⁶³ As such, **Llewellyn's** work proposes Realists are also interested in the life beyond the law as it impacts the law.⁶⁴

In analysing law, **Llewellyn** proposes behaviour as the main subject of law as an observational science since behaviour is objective.⁶⁵ Thus, the difference between what the officials say they do and what they do in fact must be an object of attention.⁶⁶ Following this, **Llewellyn** also points to the distinction between “is” and “ought”.⁶⁷ It is important to note however that a separation between “is” and “ought” does not mean that “ought” does not exist, but merely it is a different category of study. **Llewellyn** says that the question of “ought” ought to follow the question of “is”.⁶⁸ **Llewellyn** believes that Realists stir the law, and they call for attention to the interests of the practice of law.⁶⁹ His account of Realism is of Realists, the problems they deal with, the questions they seek

⁶¹ Gürkan, p. 107.

⁶² Gürkan, p. 108.

⁶³ Gürkan, p. 110.

⁶⁴ Gürkan, p. 110.

⁶⁵ Gürkan, p. 114.

⁶⁶ Gürkan, p. 114.

⁶⁷ Gürkan, p. 115.

⁶⁸ Llewellyn, 1931, p. 1223.

⁶⁹ Llewellyn, 1931, p. 1222.

answer to: “what does law *do*” and “what *ought* law do”.⁷⁰ This approach emphasizes the critical spirit in the Realist movement.

According to **Llewellyn**, law also encompasses the actions of government officials, and since people arrange their actions in accordance with the actions of these officials, the first order of business would be to find methods to predict these actions.⁷¹ **Llewellyn** does not disregard rules but puts forward the idea that traditional view of law mistakenly focuses on the words in these rules prescribing the judge to act in a certain way while disregarding the way judges act in fact.⁷²

Llewellyn says that there are three lines of attack that “converge to a single conclusion”, with the conclusion being: “there is less possibility of accurate prediction of what courts will do than the traditional rules would lead us to suppose”⁷³ and if the rules are as certain as they are claimed to be, “how can two good lawyers disagree about an appealed case?”⁷⁴

On the question of what law is, **Llewellyn** notes that he has not been concerned with defining law or marking a periphery for it.⁷⁵ As such, we must also note that Realist understanding of law by **Llewellyn** does not assign any specific meaning to the concept of law.

Llewellyn himself rolls his understanding of Realism to a single conclusion, that the rules are not certain, and they are not useful tools for predicting the outcomes of court decisions. This, in turn, also implies, as all his writings do, that it is important to be able

⁷⁰ Llewellyn, 1931, p. 1223.

⁷¹ Gürkan, p. 116.

⁷² Gürkan, p. 117.

⁷³ Llewellyn, 1931, p. 1241.

⁷⁴ Llewellyn, 1931, p. 1241n47-1242.

⁷⁵ Llewellyn, 1930, p. 464.

to predict court decisions. As we have discussed above, there are many other points to **Llewellyn** and his understanding of Realism. Still, we must realise as he has, that the Realist spirit converged in this point of origin for all their conclusions.

C. JEROME FRANK

For **Frank**, Realists simply represent a group of lawyers united in their scepticism provoked by the desire for reform in justice delivered by the courts.⁷⁶⁷⁷ **Frank** has also used the term constructive scepticism to refer to Realists and splits the Realists into two groups that he calls rule sceptics and fact sceptics.⁷⁸⁷⁹ For him, the rule scepticism refers to the ideas of those who ask for certainty in law as they see value in prediction, and they believe that they can discover the real rules behind the paper rules.⁸⁰⁸¹ On the other hand, fact sceptics believe that it is never possible to predict court decisions as the facts of a case are always variant; as such they are interested in how justice can be established in decision making processes.⁸²⁸³

Frank's main attack is towards the “basic myth” of the law, the desire to seek certainty in law and rules.⁸⁴ **Frank** also theorizes about why legal professionals might be following this basic myth.⁸⁵ **Frank's** theory also involves ideas as to the role of courts in creating law, claiming that in reality it is the courts that create law.⁸⁶

⁷⁶ Gürkan, p. 47.

⁷⁷ Frank, 1949, p. viii.

⁷⁸ Gürkan, p. 47.

⁷⁹ Frank, 1949, p. viii.

⁸⁰ Gürkan, p. 47.

⁸¹ Frank, 1949, p. viii.

⁸² Gürkan, p. 47-48.

⁸³ Frank, 1949, p. ix.

⁸⁴ Gürkan, p.49

⁸⁵ Gürkan, p.50-53.

⁸⁶ Gürkan, p. 54.

Frank's idea on what law is parallel to **Llewellyn**'s, in the sense that **Frank** also thinks that it is not possible to define law.⁸⁷ His idea in law is either past "law" that is previous decisions made by the courts or the probable "law", that is the prediction of a court decision.⁸⁸ **Frank**'s decision centric approach is also hand to hand with his idea that rules are not the law as they do not enable this business of prediction entirely on their own, due to the role of the judge in decision making.⁸⁹ **Frank**'s criticism extends to clinging to precedents and codification in search of certainty.⁹⁰

Following these ideas, **Frank** also analyses what courts do in fact and says that since rules are not always certain and due to the uncertainty in facts causing uncertainty towards the fact that which rules will be applied to a situation, court decisions cannot be predicted with certainty.⁹¹ Further, he claims that the rationale put forward by the judge for the decision cannot be trusted as the judge is a human and does not include in the conclusion that humanity.⁹²

This uncertainty, however, we must not read as an absolute uncertainty. **Leiter** notes even in the "radicalised" approach of **Frank** law is not boiled down to the gastrointestinal theory some claim that it is: "But no Realist ever advocated the gastrointestinal theory, and only one, Jerome Frank, ever came close to elevating the all-too-human psychological idiosyncrasies of individual judges to the pivotal factor in legal decisionmaking."⁹³ **Frank** claims the uncertainty lies in the variance of facts, as we have

⁸⁷ Gürkan, p. 53.

⁸⁸ Gürkan, p. 53.

⁸⁹ Gürkan, p. 56-57.

⁹⁰ Gürkan, p. 58-63.

⁹¹ Gürkan, p. 67.

⁹² Gürkan, p. 73.

⁹³ Leiter, 2015, p. 1979.

mentioned above⁹⁴, which is parallel to the idea that there may be fact-patterns as **Leiter** has noted.⁹⁵

Frank's approach is initiated with the idea that law is not certain. His analysis mainly follows from that idea, and focuses on first and foremost the need to accept this as reality. For **Frank**, it is important to be mature and accept the fact that law is uncertain.⁹⁶ His other ideas then revolve around remedies to this reality, and how lawyers can deal with this uncertainty: focusing on the human aspect of judges and the impact of their psychology.

II. NEW LEGAL REALISM

We are speaking of a renaissance of Legal Realism. This thesis predicts that Legal Realism will be reborn if legal decision making by AI models is adopted in practice. This prediction also calls for an inquiry as to New Legal Realism; as it is a Legal Realism, only new.⁹⁷ We believe New Legal Realism is mostly a consequence of Realism and argue that as New Legal Realism has diverged from Legal Realism in certain aspects and came to refer to a certain academic program, these points of divergence are not of interest to our argumentation.

For the purposes of our argument, we believe it does not matter if we consider New Legal Realism a brand-new way to do Legal Realism, or its consequence, or something else; since our argument lies in that certain tenets of Legal Realist thought are key to studying legal decision making with AI, which will result in a rebirth of Legal

⁹⁴ See *supra* note 82 and 83.

⁹⁵ See *supra* note 32.

⁹⁶ Gürkan, p. 64.

⁹⁷ For discussions regarding CLS and law and economics as a continuation of Legal Realism also see Dagan (2007) and Kalman (2021).

Realist thinking in legal scholarship. This *may* happen in a New Legal Realist fashion, or in other strands of Realism which have dropped the “realism” as a marker of their views. We argue that the tenets discussed under Section III.B. are a backbone for the Legal Realist *study* of law; and as such New Legal Realism or any other legal theory which has adopted these would be suitable as a tool in explaining the new phenomenon of legal decision making with AI. Still, in the following section; we discuss what New Legal Realism is on the surface level and how it differs from Legal Realism to provide a more solid theoretical framework.

Kalman says that New Legal Realism “translates law and social science to each other and merges qualitative and quantitative sociolegal research with the teaching, theory, and practice of law.”⁹⁸ **Mertz**’s more detailed account on this matter is as follows:

NLR arose at the turn of the millennium in the United States, in part as a reaction to the relative failure of the legal academy to pick up on social science research outside of economics, and in part in response to the rising number of social scientists with law degrees embarking on law teaching. Both in terms of translating social science for law, then, and in terms of translating law’s internal understandings for social science, the moment was ripe for a new development in a longstanding effort.⁹⁹

This ripeness is seen again in the purposes of New Legal Realism. **Meere E. Deo** says that “New Legal Realism (NLR) calls for scholars to engage with social science theories, methods, and literatures to better understand and thereby improve the law in practice.”¹⁰⁰

⁹⁸ Kalman, p. 46.

⁹⁹ Mertz, p. 28.

¹⁰⁰ Meere E. Deo, “New Legal Realism, eCRT, and the future of legal education scholarship”, *Research Handbook on Modern Legal Realism*, ed. Shauhin Talesh, Elizabeth Mertz, Heinz Klug, Elgaronline, 2021, p. 240.

Micheal McCann notes that “The core commitment of New Legal Realism to date has been summarized by its advocates as theory-driven empirical research about law in action that values qualitative as well as quantitative and experimental methods.”¹⁰¹ These aims have been realised in New Legal Realist works.

Stewart Macaulay, who is noted to be a founding member of the New Legal Realism,¹⁰² begins his account of how New Legal Realism differs from Legal Realism by saying that “the classic realists talked about doing empirical research, but relatively little was accomplished”.¹⁰³ **Thomas J. Miles and Cass R. Sunstein** also note the same difference: “The New Legal Realists are conducting what **Llewellyn** and his peers only envisioned ‘large-scale quantitative studies of facts and outcome’ that assess the influence of the judicial personality on legal outcomes.”¹⁰⁴ By this account, New Legal Realism is particularly focused on this empirical research that is the continuation of Legal Realist thought. This is also reported in **Mertz** in saying “the work of today’s Legal Realists, such as **Atuahene** and **Mitchell**, honor many aspects of what a prior generation tried to leave us as part of a Realist tradition”.¹⁰⁵

This empirical research focus is more apparent in **Miles and Sunstein** as they propose that “The relevant studies have produced a New Legal Realism—an effort to understand the sources of judicial decisions on the basis of testable hypotheses and large

¹⁰¹ Micheal McCann, “Preface to The New Legal Realism, Volumes I and II”, *The New Legal Realism, Volume I*, New York: Cambridge University Press, 2016 p. xiv.

¹⁰² Mertz, p. 28.

¹⁰³ Stewart Macaulay, “The New versus the Old Legal Realism: Things Ain’t What They Used to Be”, *Wisconsin Law Review*, Vol. 2005, No. 2, 2005, p. 375.

¹⁰⁴ Thomas J. Miles and Cass R. Sunstein, “The New Legal Realism”, *The University of Chicago Law Review*, Vol. 75, No. 2, Spring 2008, p. 834.

¹⁰⁵ Mertz, p. 33.

data sets”.¹⁰⁶ Their emphasis is on the methodologies by which they differ from “old” Legal Realism. They note this issue as follows:

A distinguishing feature of the New Legal Realism is the close examination of reported cases in order to understand how judicial "personality," understood in testable ways, influences legal outcomes, and how legal institutions constrain or unleash these influences.¹⁰⁷

They also criticize Legal Realists in saying that old Legal Realists “rested content with impressions and anecdotes, not with any kind of systematic study. By contrast, the New Legal Realists take these claims about legal reasoning as hypotheses, which can and should be tested.”¹⁰⁸

Macaulay also notes the scope of New Legal Realism, as “law in action” but in an expanded manner.¹⁰⁹ He says that this expanded meaning includes “both the gaps between the law on the books and what happened in legal institutions, and, the way problems were avoided, suppressed, and dealt with apart from official public norms, sanctions, and institutions”.¹¹⁰ In this manner, New Legal Realism proposes a scope much broader than that interested the Legal Realists at the time.¹¹¹ **Macaulay** emphasises their scope as follows: “A New Legal Realism must go beyond mere gap research. Legal pluralism is an essential idea for one who would think about law seriously. The action may be taking place quite apart from cops, administrative agencies, and courts.”¹¹²

¹⁰⁶ Miles and Sunstein, p. 830.

¹⁰⁷ Miles and Sunstein, p. 834.

¹⁰⁸ Miles and Sunstein, p. 836.

¹⁰⁹ Macaulay, p. 385.

¹¹⁰ Macaulay, p. 387.

¹¹¹ See Section III.A., notes 150 and 151.

¹¹² Macaulay, p. 388.

Macaulay also mentions that “A New Legal Realism could challenge the adequacy of studying the legal system when that concept is defined formally and narrowly.”¹¹³ **Miles and Sunstein** argue that a New Realist study of “judicial behavior will erode the distinctions between ‘law and politics’ political science and empirical legal studies”.¹¹⁴ To that end New Legal Realism goes beyond “legal” in a sense.

Macaulay also notes a difference between “tone” between the old Realists and the New ones, which he cites as follows:

At the New Legal Realism Symposium, Galanter observed that the old legal realists were optimistic and saw law as a “robust, expansive, problem-solving enterprise—by breaking the shackles of formalism, they would liberate it, unleash the power of law to solve problems—[it was] their own version of release of energy.”¹¹⁵

It seems New Legal Realism does not believe law as a means of solving problems. He notes “A New Legal Realism could tell us more about the costs and benefits of attempts to bring about change through law”.¹¹⁶

Another point by **Macaulay** is that “a new legal realism will not be able to claim to be only a disinterested, neutral, nonpartisan pursuit of the facts. Sometimes looking at the law in action will carry a political message”.¹¹⁷ His idea of a New Legal Realism encompasses political pursuit as well. This need for normative approaches is echoed in **Miles and Sunstein** as well: “But the need for a just legal system is urgent; it cannot wait

¹¹³ Macaulay, p. 399.

¹¹⁴ Miles and Sunstein, p. 834.

¹¹⁵ Macaulay, p. 391.

¹¹⁶ Macaulay, p. 392.

¹¹⁷ Macaulay, p. 392.

until researchers achieve a comprehensive understanding of the ‘Is’”¹¹⁸, urging for an ought immediately, as opposed to classical Realism.

There is also a need to examine the relationship New Legal Realism and the new AI tools. **Poppe** talks about “an opportune moment for New Legal Realism” as it relates to reorienting legal education because of new technologies:

A first reason for hope is the increasingly salient role of technology in legal practice. Current and emerging technologies ranging from e-discovery to algorithmic prediction are shaping the practice of law. Understanding these technologies, and interpreting the results they produce, requires probabilistic thinking and an appreciation for the relationship between underlying data quality and the accuracy of results. As comfort with empirical data, analysis, and inferences becomes increasingly essential to the practice of law, demand in the job market may incentivize students to obtain these skills. It could also spur top-down educational reforms consistent with the goals of New Legal Realism.¹¹⁹

This approach is quite similar to our own understanding of a renaissance in Legal Realism. **Poppe** argues that New Legal Realism’s call for educational reforms could be realised due to the increasing use of new technologies.

The educational agenda of New Legal Realism consists of the following, according to **Poppe**:

- (i) instill in students an appreciation for the value of empirical research; (ii) provide them with a framework for understanding the relationship between law

¹¹⁸ Miles and Sunstein, p. 850.

¹¹⁹ Poppe, p. 195.

and social science; and (iii) equip them with the knowledge necessary to engage critically with empirical data and analysis”.¹²⁰

These goals align with the social empirical research aspect of new Legal Realism which we have discussed above. **Poppe** emphasizes this point in differentiating New Legal Realism in saying:

Since the time of the Realists, advocates for reform have recognized the significance of students’ baseline knowledge for their ability to engage with law school materials. New Legal Realism seeks to integrate empirical scholarship into these materials...¹²¹

On top of these, New Legal Realism also seeks to integrate social science and legal doctrine in legal education.¹²² **Poppe** accepts that the currently prevalent pedagogy is counter to the New Legal Realist approaches.¹²³ This is in line with our argument that Legal Realism has fallen from grace, a fate shared by its new form.

Above, we have discussed differences in (1) actual research done and methodologies, (2) scope of research, (3) outlook towards the effectiveness of law, and (4) the temporary separation of “is”es and “ought”s. New Legal Realism, as a continuation of Legal Realism takes it a step forward in research and actually does empirical studies. It broadens the scope of it in the same spirit of understanding legal apparatus as it impacts humans and moves beyond disciplinary bounds. In a more radical manner, it distrusts the whole of legal establishment and how it would be beneficial for certain purposes, and they ask for purposes normatively.

¹²⁰ Poppe, p. 196.

¹²¹ Poppe, p. 196.

¹²² Poppe, p. 198.

¹²³ Poppe, p. 195.

In considering these, New Legal Realism aims to follow through with the Legal Realist purposes in study. In this manner, it is a consequence of Legal Realist ferment. On the other hand, its wider scope that is in touch with political science, its outlook towards the law and normativity deserves its own place in research. For these reasons, we have considered new Legal Realism as beyond the scope of our discussions. However, we need to echo the idea that adoption of new technologies being an opportune moment for New Legal Realism in terms of educational agenda, which runs parallel to our arguments in this study. While we are focused on understanding legal decision making, the educational aspect of the Realist spirit, carried on by New Legal Realism, is no less important.

III. LEGAL REALISM IN PARTICULAR

*An hour in Chrysalis to pass,
Then gay above receding grass
A Butterfly to go!
A moment to interrogate,
Then wiser than a "Surrogate,"
The Universe to know!*¹²⁴

With having summarised **Frank** and **Llewellyn** and Legal Realist thought and New Legal Realism in general, this section will move on to our outlook on Realism. We will first answer the question of what we mean by “law”. This is important as it will frame Realism in our vision as a Realism towards the study of law, what the study of law ought to do. In order to do that, we will follow Realist discussions from **Holmes** to **Llewellyn** and **Frank**. After which we will try and answer what “Realism” entails for this thesis, as limited to our scope. Since the approach of this thesis is one that investigates legal decision making by AI and how Realism might work as a theoretical framework around

¹²⁴ Dickinson, p. 60.

it; we first explain what Legal Realists think about legal decision making process, and then move forward to points of study of Legal Realism as they relate to our subject.

In trying to lay out our outlook we do not claim that we have the one true outlook on Legal Realism, only one that makes sense for our scope. To quote **Llewellyn**: “What realism was, and is, is a method, nothing more, and the only tenet involved is that the method is a good one. ‘See it fresh,’ ‘See it as it works’ – that was the foundation of any solid work, to any end.”¹²⁵ This reflects our approach as well. On top of that, **Dagan** also notes a similar point about his views on Realism:

Indeed, I am not claiming to have discovered the one true essence of legal realism that is obvious to anyone who cares to read the legal realist literature. The reading suggested here is unashamedly influenced by my own convictions about law. Nor am I pretending, however, to have invented the ideas I present.¹²⁶

The ideas we present here are not new, they are not invented, and they cannot be claimed to be the one true essence of Realism. However, they reflect opinions on Realism refined for the purposes of our study, and informed by the Realist literature as we discussed in length and guided by a Realist outlook.

Brian Bix notes that “the “essence” and the “legacy” of American Legal Realism remains highly contested”.¹²⁷ **Tamanaha** again criticizes trying to pinpoint attributes of Legal Realists:

¹²⁵ Karl N. Llewellyn, *Common Law Tradition: Deciding Appeals*. Boston, Toronto: Little Brown, 1960, p. 510. Heinonline.

¹²⁶ Dagan, p. 609.

¹²⁷ Brian Bix, The Many Faces of Modern Legal Realism, *Revista Ordines*, Vol. 2022/2, 2022, p.35.

[...] interpretations can point to several examples among the legal realists, but none can fairly be attributed to all twenty individuals identified by Llewellyn. Statements about what the legal realists stood for, as Twining has repeatedly argued, are prone to overgeneralization: “What is clear is that the longer the list, the harder it is to make accurate, precise, and nontrivial general statements about Realism or the Realist”¹²⁸

It is not that we seek to find a Realism to fit all, nor is it that we are seeking *the* essence of Realism; but we are seeking *an* essence of Realism, as many have before. To this end, as we have discussed above, we will first try to lay out our understanding for what law refers to in Realist discussion, at least for our scope. After that, we will discuss what we understand by Realism and what it refers to in the context of this study.

A. WHAT LAW REFERS TO

In seeking a Realist vision, it is important to note that thinkers who can be or were labelled as Realists expressed differing opinions. **Llewellyn**, at the time, analyses twenty different authors and approximately ninety titles as the Realist ferment.¹²⁹ What “unites” these people as Realists is not necessarily the answers which they have proposed to the questions in law but rather their approach, as **Llewellyn** points out:

They want law to deal, they themselves want to deal, with things, with people, with tangibles, with *definite* tangibles, and *observable* relations between definite tangibles — not with words alone; when law deals with words, they want the words to represent tangibles which can be got at beneath the words, and observable relations between those tangibles. They want to check ideas, and rules

¹²⁸ Tamanaha, p. 161.

¹²⁹ Llewellyn, 1931, p. 1227.

and formulas by facts, to keep them close to facts. They view rules, they view law, as means to ends; as only means to ends; as having meaning only insofar as they are means to ends. They suspect, with law moving slowly and the life around them moving fast, that some law may have gotten out of joint with life.¹³⁰

For **Llewellyn** these people are grouped together not because their likeliness in their belief or work but because they have departed from common points and there has become an unplanned and unforeseen whole.¹³¹

We have highlighted the thoughts of **Llewellyn** and **Frank** in the previous chapter, **Frank**, for example, proposed psychology as an end to the line of questioning brought about forward by Realism. This section will contain our understanding of this spirit of Realism, the line of inquiry rather than the various answers proposed by numerous Realists. In the beginning of this thesis, we have noted that we do not intend to define law or deal with definitions of law. We have also noted **Llewellyn** and **Frank** also were theorising in a similar manner, trying to avoid definitions. It is, however, important to make the concepts we are talking about clear as to avoid confusion regarding our arguments.

Frank, distinguishes the duty of the law school as the teaching of four separate arts, namely; (a) “the art of the judge”, (b) “the art of the lawyer”, (c) “the art of the critic of the work of lawyers and judges”, (d) “the art of the teacher of the other three arts”.¹³² **Llewellyn** also distinguishes what he understands from law, on the one hand an observational science, on the other an art of living, and finally a philosophy or a collection

¹³⁰ Llewellyn, 1931, p. 1223.

¹³¹ Llewellyn, 1931, p. 1235.

¹³² Jerome Frank, “Are Judges Human? Part Two: As through a Class Darkly”, *University of Pennsylvania Law Review and American Law Register*, Vol. 80, No. 2, Dec. 1931, p. 259.

of norms or social objectives; and it also encompasses the rules.¹³³ Even further so, **Frank** proposes that whenever the word law is used the user put the intended meaning in parentheses as to avoid confusion.¹³⁴ In this study, we are instead devoting a section towards this discussion, to show what we mean by law and why we mean that.

With these distinctions in mind, we can examine some remarks as to what law is. In a sermon preached before the King, the **Bishop of Bangor, Benjamin Hoadly** has put forward the proposition: “whoever hath an absolute authority to interpret any written, or spoken laws; it is he, who is truly the law giver, to all intents and purposes; and not the Person who first wrote, or spoke them.”¹³⁵ In adopting a decision centric approach to Law, the study of Law becomes a study of decisions and how they come to be. **Holmes** echoes this in “The Path of the Law”: “The object of our study, then, is prediction, the prediction of the incidence of the public force through the instrumentality of the courts.”¹³⁶ Thus, a law is a prophecy as **Holmes** calls them, a probable outcome as to what the decision will be. Again, as **Holmes** puts it, the law to keep a contract only means that you might have to pay damages if you do not keep it.¹³⁷

In strictly following **Holmes**, the argument would be that a lawyer’s duty is to predict decisions, and laws are prophecies which would aid them in this cause. It must be noted, however, that there has to be made a distinction: if laws are *described* as such then there has to be other qualities which make laws, separating them from other prophecies

¹³³ Gürkan, p. 113.

¹³⁴ Frank, Are Judges Human Part One, 1931, p. 45.

¹³⁵ Benjamin Hoadly, *The nature of the kingdom or church, of Christ: A sermon preach’d before the King, at the Royal chapel at St. James’s, on Sunday March 31, 1717. By Benjamin, Lord Bishop of Bangor*, London: printed for James Kanpton, and Timothy Childe, Eighteenth Century Collections Online Text Creation Partnership, 1717, p. 12.

¹³⁶ Oliver Wendell Holmes, “The Path of the Law”, Harvard Law Review, Vol. 12, No. 7, Feb. 25, 1897, p. 457.

¹³⁷ Holmes, p. 462.

regarding decisions; but if laws are *defined* as such then any prophecy regarding the decision of a court is a law.

Holmes appears to be *defining* the law in calling: “The prophecies of what the courts will do in fact, and nothing more pretentious, are what I mean by the law.”¹³⁸ This definition, however, does not give a distinction as to what is the law and what are the sources of law, and with good reason; for **Holmes**, there is no need for one:

You may assume, with Hobbes and Bentham and Austin, that all law emanates from the sovereign, even when the first human beings to enunciate it are the judges, or you may think that law is the voice of the *Zeitgeist*, or what you like. It is all one to my present purpose.¹³⁹

This reiterates his pragmatic approach: **Holmes** is not interested in a limitation; he is only interested in the ability to predict the consequences of actions brought to a court, an idea later echoed by **Frank**.¹⁴⁰

The approach focusing on predictions of court decisions and defining all prophecies regarding what the courts will do as law invokes a rule-nihilist approach to the nature of the study. There are no rules because those rules are not certain to be followed; there are simply inclinations, or trends. Therefore, the law is not the “rule” in the criminal code that tells us “when a person kills another, they are penalised”; but it is that prediction which claims “usually, when a person kills another, the consequence is that they are penalised.”

¹³⁸ Holmes, p. 461

¹³⁹ Holmes, p. 465

¹⁴⁰ Gürkan, p. 56.

Walter Whaler Cook approaches the issue of law in a similar vein, emphasising an inductive methodology in the generation of rules or principles:

The actual process involved in settling a situation of doubt—a new case, if we are dealing with law—involves a comparison of the data of the new situation with the facts of a large number of prior situations which have been subsumed under a “rule” or “principle” within the terms of which it is thought the new situation may be brought.¹⁴¹

It is important to note that this view differs from **Holmes** in the sense that it recognizes there are rules however they may come to be. **Cook**, here, does not seem to be interested in what law is but how law is practiced, still, he is parallel to **Holmes** in the sense that his outlook involves an application of previous data to the new situation. This would of course require an inquiry as to what the courts did do, in fact.

Judge **Hutcheson**, in “The Judgment Intuitive: The Function of the ‘Hunch’ In Judicial Decision” reiterates this idea through emphasizing the role of intuition in the way judges decide cases. **Hutcheson** describes the process as following a hunch and proposes that he first feels his way to a judgment through the use of his hunch and works backward in inventing an opinion.¹⁴² **Hutcheson** draws to comparison to use of dice and claims that this use of dice is what drives humanity to solutions not only in Law.¹⁴³

Llewellyn tries to clarify the argument by saying that he “is not arguing that the ‘rules of substantive law’ are without importance”, instead what he is actually arguing is that there are less important than they are assumed to be and they should be removed from

¹⁴¹ Walter Whaler Cook, “The Logical and Legal Basis of the Conflict of Laws”, *The Yale Law Journal*, Vol. 33, No. 5, Mar. 1924, p. 486.

¹⁴² Joseph Chappell Hutcheson Jr., “Judgment Intuitive The Function of the Hunch in Judicial Decision”, *Cornell Law Review*, Vol. 14, Issue 3, April 1929, p. 280.

¹⁴³ Hutcheson, p. 278-279.

the focal point of legal discussion.¹⁴⁴ He then emphasises that it is assumed [incorrectly] that the practices of the judges are in conformity with the oughts in the books and these oughts describe the is-es.¹⁴⁵

Llewellyn makes an important distinction apparent. Law as a concept that is the heart of the discussion and Law as the study of that concept are two different issues. This is the distinction we have touched upon at the beginning of this discussion. The ideas **Holmes** had put forward as to what the law is had to change what the study of Law meant as well. If Law is of words, then its study is a study of words; if Law is of actions, then its study is a study of actions. However, Law; referring to so many concepts at once creates a conflict of meanings: (1) on one hand we have the law in the “orthodox” sense, that is law in the “narrow” sense that refers to the set of rules; the “oughts” in the books that *impact* the outcomes in courts, (2) on the other we have the law in the broader sense, the object of the Realist study, “what the courts do, in fact” (though, the scope may vary), (3) and there is Law as an area of study.

Llewellyn’s call appears to be a call for an emphasis change, in the *study of the law*. This call can be interpreted as simply the part of a pragmatist epistemology agenda; the study of the law *ought to* be predictive, and thus must be a study of what the courts do. This is also compatible with the question-oriented understanding of Realism, as the study of law refers to the questions that ought to be asked.

This problem of confusion regarding the concept of Law is put forth by **Frank** as well, in “What Courts Do In Fact”; **Frank** proposes simply that it is better to answer the question of what courts do in fact instead of what law is.¹⁴⁶ Both **Frank**, and

¹⁴⁴ Llewellyn, 1930, p. 442.

¹⁴⁵ Llewellyn, 1930, p. 443.

¹⁴⁶ Jerome Frank, “What Courts Do in Fact – Part One”, *Illinois Law Review*, Vol. 26, No. 6, 1931-1932, p. 645.

Llewellyn redefine the *study of the law*, they redirect all the questions for their one pragmatic purpose. It is paramount to see how **Llewellyn** and **Frank** rescue this discussion from being a mere play of semantics, the question as to what the law or Law means will inevitably be purely an issue of definition. The Realist question, as it relates to our scope, is twofold in this sense: (1) what the *study of the law* ought to be, the scope of it; (2) and what that study will show, what will be the answers to the questions of issues. **Llewellyn** is therefore diverging from **Holmes**¹⁴⁷ and frames the Realist vision.

We have tried to answer the question what we refer to as law and established that Realism does not simply deal with an abstract concept of law but instead focuses primarily on how to study law in practice. We did this in order to clear up any confusion relating to our understanding of Realism. With their ties to rule-scepticism and their decision centric approach, it is quite easy to misinterpret the Realist claims as a redefinition of what law is. However, on the contrary, Realism abstains from defining law, and focuses on observing it.

In choosing to observe certain phenomena and disregarding others, Realists *de facto* declare a space for law, that is true. This scope is made issue in his foreword to *The Bramble Bush* by **Llewellyn** as well. **Llewellyn** discusses his own remark about law: “what these officials do about disputes is, to my mind, the law itself.”¹⁴⁸ He says that this single sentence has been used to characterise him, and still, there is more to matter than that.¹⁴⁹ **Cook** and **Frank** who respectively argued that legal study should encompass

¹⁴⁷ and Hoadly, but he is not to be considered part of the realist thought for our purpose, simply a prophecy of it.

¹⁴⁸ Karl N. Llewellyn, “Foreword”, *The Bramble Bush*, Louisiana: Quid Pro Books, 2012, p. 7-11.

¹⁴⁹ Llewellyn, 2012, p. 7-11.

officials and all officials concern lawyers.¹⁵⁰¹⁵¹ Yet, these should be read together with the ideals these authors have presented before and not as a way to define and limit what law encompasses but as a way to limit the scope of their discussions. **Llewellyn's** reproach here is not unfounded, in picking an area of study, Realists do not aim to define law, but only to define a scope of work.

After establishing the Realist ideas around law and the study of it, we need to understand what Realism means for the scope of this thesis. With this in mind, in the next section we will talk about the Realist ideas relevant to our scope. These mostly will be summaries of the ideas we have discussed above.

B. WHAT LEGAL REALISM ENTAILS

In this section we are first going to examine the role of the judge and the process of legal decision making in Realism. There is discussion to be made what the judge ought to do. Judge is often expected to provide justice to both sides. There are legal duties prescribed by rules to the judges as well. However, as we have established, we will focus on not what the judges ought to do but what they do, in fact.

Realists do not deny the judges their justice, they only realise that justice as an is, that judges seek to establish justice and they do so; and they question the methods by which that happens. **Roscoe Pound** says, “Courts could not forget that they were administering justice, and the most that such a theory could do was to hamper the judicial instinct to seek a just result”¹⁵² in “The Theory of Judicial Decision”, as a point of

¹⁵⁰ Lon L. Fuller, *The Law in Quest of Itself*, Boston: Beacon Press, 1966, p. 52n12.

¹⁵¹ Fuller, p.53n13.

¹⁵² Roscoe Pound, “The Theory of Judicial Decision. III. A Theory of Judicial Decision for Today,” *Harvard Law Review*, Vol. 36, No. 8, 1923, p. 940.

criticism towards any theory that denies the judicial instinct of justice present within the judge.

Pound then moves on to explain the steps a judge takes to decide a case. He mentions three steps in the process of thinking for the judge: (1) “selection of the legal material on which to ground the decision, or as we commonly say, finding the law”; (2) “development of the grounds of decision from the material selected, or interpretation”; and (3) “application of the abstract grounds of decision to the facts of the case”.¹⁵³

Pound argues that these three steps are all worked with intuition.¹⁵⁴ It is the Realist stand that the judge does these not by any precise faculty of the mind but through the hunch that has been developed through experience. **Pound**’s explanation is as follows:

One is in the selection of grounds of decision - in finding the legal materials that may be made both to furnish a legal ground of decision and to achieve justice in the concrete case. It is an everyday experience of those who study judicial decisions that the results are usually sound, whether the reasoning from which the results purport to flow is sound or not. The trained intuition of the judge continually leads him to right results for which he is puzzled to give unimpeachable legal reasons. Another place where the judge's intuition comes into play is in development of the grounds of decision, or interpretation. This is especially marked when it becomes necessary to apply the criterion of the intrinsic merit of the possible interpretations. A third is in application of the developed grounds of decision to the facts.¹⁵⁵

¹⁵³ Pound, p. 945.

¹⁵⁴ Pound, p. 951.

¹⁵⁵ Pound, p. 951.

While **Pound** argues that there are steps to the judgement, he does emphasize that the judge goes through these steps with intuition.

Frank also notes the conventional description in a two-step manner that is (1) determination of the facts (which **Pound** takes as given) and (2) the determination of what rules should be applied to those facts.¹⁵⁶ Still, **Frank**'s approach is arguably more radical compared to **Pound**. **Frank** argues that nobody knows how judges make decisions.¹⁵⁷ His approach supposes that the determining factors in decisions are not facts or rules, which are still present in **Pound**, but the Stimuli and Personality in what he formulates as "S X P = D"¹⁵⁸. **Frank** does not deny that rules or facts do play a role in the judgement in saying that "The so-called legal rules and principles are some of the many hunch producers".¹⁵⁹ Yet, he still argues that the stimuli and the personality of the judge are the main determining factors in the decision, as they are what these hunch producers operate through.¹⁶⁰

Max Radin also suggests that judges are human and thus will act like human beings.¹⁶¹ As such, he voices distrust in the conventional thinking as to how judges decide cases:

Now what does His Honor do? Does he search his mind for the principles of justice, divide them into classes I, II, and III, and under class III, "Duties to one's Fellow-man" does he subdivide into (a), Duties of Active Benevolence, etc., until in this process of division and subdivision, classification and analysis he comes

¹⁵⁶ Frank, 1949, p. xi.

¹⁵⁷ Jerome Frank, "What Courts Do in Fact – Part Two", *Illinois Law Review*, Vol. 26, No. 7, 1931-1932, p. 773.

¹⁵⁸ Frank, "What Courts Do in Fact – Part Two", 1931-1932, p. 775.

¹⁵⁹ Frank, "What Courts Do in Fact – Part One", 1931-1932, p. 662.

¹⁶⁰ Frank, "What Courts Do in Fact – Part One", 1931-1932, p. 662.

¹⁶¹ Max Radin, "The Theory of Judicial Decision: Or How Judges Think", *American Bar Association Journal*, Vol. 11, No. 6, 1925, p. 357.

upon something that will exactly fit such qualified human beings as green grocers and the officials of a street railway company. Somehow, I do not believe he does.¹⁶²

The distrust by **Radin** is shared by **Frank** as well, for **Frank** judges do not sincerely explain how they decide and conceal the role of their personality.¹⁶³ What must be considered is not what judges say they do, but what they do.¹⁶⁴

Llewellyn notes the angle of attack the Realists had adopted against this distrust towards how decisions are made: rationalisation.¹⁶⁵ Following **Hutcheson** he reiterates the perspective which views decision as a backwards process by Realists through rationalization:

To recanvass the opinions, viewing them no longer as mirroring the process of deciding cases, but rather as trained lawyers' arguments made by the judges (after the decision has been reached), intended to make the decision seem plausible, legally decent, legally right, to make it seem, indeed, legally inevitable— this was to open up new vision.¹⁶⁶

Judges such as **Hutcheson** called for recognition of the role of hunches in legal decision making. These hunches act as the devices of those who cannot make infinitely accurate calculations, as per the limitations of our cognition. We cannot “find” the “true meanings” of laws before us, we cannot decide whether a general rule applies to a specific case in certainty, as such we must adapt and learn to follow “intuition”.

¹⁶² Radin, p. 357.

¹⁶³ Gürkan, p. 58.

¹⁶⁴ Gürkan, p. 59.

¹⁶⁵ Llewellyn, 1931, p. 1238.

¹⁶⁶ Llewellyn, 1931, p. 1238.

Hutcheson recounts his employment of hunches and promotes it as “that tiptoe faculty” enabling much innovation.¹⁶⁷ **Frank** also quotes in “What Courts Do in Fact” the words of **Chancellor Kent**: “I saw where justice lay, and the moral sense decided the court half the time. I then sat down to search the authorities ... I might once in a while be embarrassed by a technical rule, but I almost always found principles suited to my view of the case.”¹⁶⁸ This is how judges act, following hunches and heuristics, in fact. This awareness is also present in **Cardozo, Hand, Lehman, Holmes, Hutcheson**, and others according to **Frank**.¹⁶⁹¹⁷⁰ To quote **Zaky**, “Realists have repeatedly stressed the role of the judicial hunch in aiding the judge in his or her legal decision-making. With the majority of Realists recognizing its central role, some differ as to what particular elements define or determine said hunch.”¹⁷¹

Through these discussions, we come to see that Realists approach the judicial decision making with a lot of scepticism towards the traditional understanding and accounts. They consider the role of rules secondary and put focus in the hunches of the judge and how those are shaped. They do consider the process of decision making as steps, with the determination of facts at the beginning and then the steps explained by **Pound** following. Still, they do not rule out that sometimes judges to reach a decision and consider the facts and rules available to them with that decision in mind. That decision might be, and might ought to be, shaped by the intentions such as fairness and justice.

The Realist view towards legal decision making relates to our study because we will investigate how it compares to legal decisions put forward by AI; however, our

¹⁶⁷ Hutcheson, p. 279.

¹⁶⁸ Frank, “What Courts Do in Fact – Part One”, 1931-1932, p. 654.

¹⁶⁹ Frank, 1949, p. 153.

¹⁷⁰ Gürkan, p. 60.

¹⁷¹ Zaky, p. 50.

approach is more leaning on how AI might be studied with Realism, as opposed to how AI will compare to human judges. As such, we must determine antecedents to the Realist approach. With that, we agree with **Leiter** on the following: “[Realists] may not give us paradigms of good empirical studies, but perhaps we should not look to them for that. The Realists give us the philosophical motivation and cues for how we should proceed, even if they do not carry off the project themselves.”¹⁷²

Maybe the first and foremost, and maybe the only postulate of Legal Realism as it relates to our thesis is their claim that law is uncertain. Realists diverge from the rest of their contemporaries by the virtue of their line of questioning that arises out of the most fundamental problem that they have encountered with law; the fact that law is uncertain, at least less certain than a traditional rules-based approach would suggest.¹⁷³¹⁷⁴ This will be the first point for our understanding of Legal Realism. This idea is reflected again in **Kalman** as follows: “[Legal Realism] reflected an attempt to increase law’s predictability by first acknowledging that legal rules, as traditionally derived, did not guarantee legal certainty”.¹⁷⁵

As we have mentioned before, our understanding of Realism stipulates that Realism is concerned with the study of law. As such, our following points of Realism will be emanating from this postulate and will be ought statements towards the study of law. That is, the study of law as it relates to the practice of law.

The next important point is that Realists were primarily concerned with law in effect. Realists studied law in action, how lawyers predicted cases and how judges decided them. This is hand to hand with the pragmatic approach that is present in the

¹⁷² Leiter, 1997, p. 312.

¹⁷³ Llewellyn, 1931, p. 1241.

¹⁷⁴ Frank, 1949, p. 5.

¹⁷⁵ Kalman, p. 39.

Realism. Realism is concerned with useful information that is available to the one who studies the legal phenomena and argues for the use of any tool. **Frank** and **Llewellyn** has both made points about the use of psychology in understanding judges, as judges were their subject and psychology would provide useful tools to understand them.

The third point would be the removal of rules from the focal point of legal study.¹⁷⁶ This follows naturally from the postulate as the uncertainty of law also implies the uncertainty of rules, and uncertain rules cannot remedy the uncertainty of law on their own. Therefore, the study of law must study more than the rules. This is stated by **Llewellyn** as “distrust of the theory that prescriptive rule-formulations are the heavily operative factor in producing court decisions”.¹⁷⁷

The fourth point that considers us is the “is” and “ought” separation in legal study. This point is highlighted by **Leiter**, “Realists want to understand and explain the way things really are, rather than engage in idle talk about how they "ought" to be”¹⁷⁸. To this end, Realists have stood behind the idea that “is” and “ought” in law should be separate from each other, as we would have to understand what “is” in order to be able to strive for what ought to be.¹⁷⁹ This is worded by **Llewellyn** as “the *temporary* divorce of Is and Ought for purposes of study”.¹⁸⁰

Our last point is adherence to interdisciplinary research, particularly psychology in understanding judicial behaviour. This has been pointed out by **Gürkan** and is apparent in both **Llewellyn** and **Frank**’s writings. This is an end to the means of understanding judges, and Realists aimed to not shy away from non-traditional tools in studying the

¹⁷⁶ Llewellyn, 1930, p. 442.

¹⁷⁷ Llewellyn, 1931, p. 1237.

¹⁷⁸ Brian Leiter, “Classical Realism”, *Philosophical Issues*, Vol. 11, Social, Political, and Legal Philosophy, 2001, p. 245.

¹⁷⁹ Gürkan, p. 38.

¹⁸⁰ Llewellyn, 1931, 1236.

subject of their study. **Leiter** echoes this inter-disciplinary approach towards the social sciences, while echoing the descriptive nature of Realists in saying, “Realists like Oliphant—who were, to repeat, the vast majority— thought that the task of legal theory was to identify and describe—not justify—the patterns of decision; the social sciences were the tool for carrying out this nonnormative task.”¹⁸¹

We have established that while we cannot point to a Realist school of thought, Realists converged around a divergence from the traditional legal theory: they believed that certainty of rules and the role of rules in court decisions are to be questioned, and through a pragmatist approach, that rules should not be considered the focal point of legal study. In this regard, they create space for all other matters to be included in legal studies. In the next chapter, we explore the current AI models used in legal decision prediction and how they work.

¹⁸¹ Leiter, “Legal Realism and Legal Positivism Reconsidered”, 2001, p. 282.

CHAPTER 2:

THE SILICON ORACLES

There are currently AI models used in legal decision predictions. When we consider the success of these prediction machines in emulating human judges, we believe there is a chance that they might be used in a legal decision making setting in the future, on their own. In a sense, if a machine is able to predict how a human would classify a case then it can (not necessarily should) be used to classify that case without human supervision. In order to understand if the decision process by these machines can be understood by legal realism, we first need to understand how they work as well. As such, in this chapter, we try to explain how these AI models work as far as they concern our scope.

The Oracle of Delphi, Pythia, was the name given to priestesses dedicated to Apollo.¹⁸² The oracle could be consulted on the seventh day of each month through the spring.¹⁸³ During these days, the oracle would bathe in a spring and burn offerings to Apollo while the priests of the temple sprinkled water on a goat to see if the goat shuddered, ensuring the favour of Apollo for the consultation to proceed.¹⁸⁴ Consultants also had to be purified, get into a queue, and pay a considerable amount of fee ensuring the consultation was necessary.¹⁸⁵ When the time came, a consultant would enter the temple, make a sacrifice.¹⁸⁶ Pythia would speak of her prophecy from a tripod as a delightful fragrance filled the chambers.¹⁸⁷ There is debate on the source of Pythian prophecies, there are arguments that the geology of the location of the temple allowed

¹⁸² Micheal Scott, *Delphi: A History of the Center of the Ancient World*, New Jersey: Princeton University Press, 2014, p.12.

¹⁸³ Scott, p. 13.

¹⁸⁴ Scott, p. 14-15.

¹⁸⁵ Scott, p. 17.

¹⁸⁶ Scott, p. 17.

¹⁸⁷ Scott, p. 20.

anaesthetic gases to induce a trans-like state in Pythia and thus the oracle had prophesised.¹⁸⁸

An oracle does not have to consult reason or logic. Differing definitions has been previously proposed for what artificial intelligence means, which can be summarized as “systems that think like humans”, “systems that think rationally”, “systems that act like humans”, and “systems that act rationally”.¹⁸⁹ The current connotations of artificial intelligence systems have swayed from syllogism, reminiscent of oracles, and have adopted architectures that act almost heuristically, as we cannot precisely define the tasks we demand formally in a step-by-step structure due to the complexity of their nature.

The complexity of deciding a case -or predicting the outcome of one- is one of the main subjects of the entire discipline of law. Even in the discipline of law, there are severely differing opinions as to what happens in a court. As a result, it is almost impossible to create a system with clear instructions as to how to decide a case. It could be argued that the rules of the legal system are a set of instructions to humans, guiding them on how to decide cases. However, it must be noted that laws are general, and we have previously established the difficulties pertaining to applying a general law to a specific case. Thusly, current legal AI systems must adopt different paths than simple syllogism.

As noted in **Zaky**, “AI is currently being used in a multitude of ways; most notably, drafting contracts, judicial prediction, arbitrator selection, determining recidivism rates and identifying legal precedent”.¹⁹⁰ However, we are interested in models

¹⁸⁸ Jelle Z. de Boer, “The Oracle at Delphi: The Pythia and the Pneuma, Intoxicating Gas Finds, and Hypotheses”, *Toxicology in Antiquity*, 2019, p. 147-148.

¹⁸⁹ Stuart J. Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach*, New Jersey: Prentice Hall, 1995, p. 5.

¹⁹⁰ Zaky, p. 29.

used in legal decision prediction, as almost “primal” versions of potential legal decision making models. We believe understanding how these models work is important in understanding how legal theory might apply to them. As we have discussed above with the Pythia, these models are almost black box models, which we believe emphasizes our need to theorise about them if we are to theorise about how they might come to make legal decisions.

In this chapter, we have noted some of the work done in legal decision prediction and underlying models used in these works. We have put forward especially the data used in these models and how they used this data. We have also noted their prediction scores reflecting their ability to mimic human judges in results (not necessarily in methods) as we believe this signals incentives to utilise these in the future instead of human judges. The first section in this chapter deals with the models in groups while the second section deals with their common points.

I. COURTROOM PREDICTIONS TODAY

The idea that computers could help with the legal profession in the form of predicting decisions is not remotely new.¹⁹¹¹⁹² Today, there is a plethora of research dedicated to this idea. Some of the literature has been surveyed in 2018 by **Ilias Chalkidis** and **Dimitrios Kampas** to the deep learning extent,¹⁹³ and there has been even more research in the following quinquennium. The scope of this thesis, however, revolves

¹⁹¹ Reed C. Lawlor, “What Computers Can Do: Analysis and Prediction of Judicial Decisions”, *American Bar Association Journal*, Vol. 49, No. 4, April 1963, p. 337.

¹⁹² Bruce G. Buchanan and Thomas E. Headrick, “Some Speculation about Artificial Intelligence and Legal Reasoning”, *Stanford Law Review*, Vol. 23, No. 1, Nov. 1970. p. 40.

¹⁹³ Ilias Chalkidis and Dimitrios Kampas, “Deep learning in law: early adaptation and legal word embeddings trained on large corpora”, *Artificial Intelligence and Law*, 2018.

around a simpler question: what are the main methods of legal prediction today, and how do they relate to each other?

A paper aiming to predict outcomes of cases decided by Turkish higher courts has been published by **Emre Mumcuoğlu and others** in 2021, which has compared its results to other papers trying to achieve similar objectives in different jurisdictions but with also different methodologies.¹⁹⁴ The survey of literature includes the following methodologies: decision trees (DT) and random forests (RF), support vector machines (SVM), and deep learning (DL) approaches.

Through this section, each method here will be broken down as to how they approach the prediction question and how they work in general. This analysis is thus limited to decision prediction tasks and will try to first give a basic understanding of each methodology and then how they were used to predict decisions. It is, however, necessary to note that the problem of predicting a case is often synonymous with classifying a case, and thus these terms will be used interchangeably throughout this section.

A. DECISION TREES

Decision trees are, in essence, a sequence of tests where a value is compared against other values.¹⁹⁵ Their advantage is often considered to be that they are more comprehensible as each node they contain can be interpreted more easily by humans.¹⁹⁶ Decision Tree architecture, as applied to legal prediction problems is used in a classification setting; that is, it is to be used when there are a given set of measurements

¹⁹⁴ Emre Mumcuoğlu et al., “Natural language processing in law: Prediction of outcomes in the higher courts of Turkey”, *Information Processing and Management*, Vol. 58, 2021.

¹⁹⁵ S.B. Kotsiantis, “Decision trees: a recent overview”, *Artificial Intelligence Review*, Vol. 39, No. 4, 29 June 2011, p. 1.

¹⁹⁶ Kotsiantis, p. 1.

on a case or object, and the aim is to find a systematic way of predicting what class it is in.¹⁹⁷

A decision tree approach to predicting US Supreme Court Cases has been adopted by **Ruger et al.**, comparing the decision tree approach to legal expert predictions.¹⁹⁸ Their statistical model consisted of only six variables (which were retained out of ten originals since the other four had no predictive power), namely: (1) “circuit of origin”, (2) “issue area of the case”, (3) “type of petitioner”, (4) “type of respondent”, (5) “ideological direction of the lower court ruling”, and (6) “whether the petitioner argued that a law or practice is unconstitutional”.¹⁹⁹ The authors also mention that they have adopted this method because of its transparency.²⁰⁰ Their model consisted of eleven decision trees at the end²⁰¹ and had achieved 75% accuracy.²⁰²

The decision tree method, in its core involves labelling a lot of data, 628 cases in the case of **Ruger et al.** study.²⁰³ On the other hand, **Mumcuoğlu et al.** has automatically extracted features from their data²⁰⁴ and have achieved roughly 70% balanced accuracy score amongst their overall corpus when they utilised decision trees.²⁰⁵

Random forests are examined alongside decision trees simply because a random forest is defined as a collection of such trees in a certain fashion.²⁰⁶ This relatively new method was used by **Micheal Benedict L. Virtucio and others** and **Daniel Martin Katz**

¹⁹⁷ Leo Breiman et al., *Classification and Regression Tress*, Chapman & Hall/CRC Press, 1998, p. 3.

¹⁹⁸ Ruger et al., p. 1161.

¹⁹⁹ Ruger et al., p. 1163.

²⁰⁰ Ruger et al., p. 1164.

²⁰¹ Ruger et al., p. 1165.

²⁰² Ruger et al., p. 1167.

²⁰³ Ruger et al., p. 1161.

²⁰⁴ Mumcuoğlu et al., p. 8.

²⁰⁵ Mumcuoğlu et al., p. 12.

²⁰⁶ Breiman et al., p. 6.

and others in predicting Philippine Supreme Court and Supreme Court of the United States respectively. **Katz et al.** used an already existing dataset for the cases which included 200 years of data and 240 values.²⁰⁷ They have cited the wisdom of the crowds as a reason for success of the random forest method. In their explanation their algorithm works by having multiple trees created by using bootstrap aggregation, a method in which random rows are selected from a tree to create new trees, and then this crowd of trees output a result that represents a better forecast.²⁰⁸ Using random trees, **Katz et al., Muhammed Burak Görentaş, Engin Yıldırım and others** and **Mumcuoğlu et al.** manage to reach roughly 70% accuracy,²⁰⁹²¹⁰²¹¹²¹² while **Virtucio et al.** reaches 59% accuracy rate.²¹³

Considering the way decision trees and random forests work, the six parameters of **Ruger et al.** are interesting, as they show that legal prediction task does not necessarily rely on legal analysis. None of their variables were regarding an area of law. On top of that, this analysis is based almost purely on statistics, still, providing a better accuracy than the legal experts who only managed a 59.1% score,²¹⁴ which is almost only as good as the worst result here by **Virtucio et al.** which had troubles due to the lack of structure in the Philippine Supreme Court cases.²¹⁵

²⁰⁷ Daniel Martin Katz et al., “A general approach for predicting the behavior of the Supreme Court of the United States”, *PLoS ONE*, 12(4), 12 April 2017, p. 4.

²⁰⁸ Katz et al., p. 7.

²⁰⁹ Katz et al., p. 8.

²¹⁰ Muhammed Burak Görentaş, *Yapay Zekâ Yöntemleriyle Uyuşmazlık Mahkemesi Kararlarının Tahmini*, Bursa Uludağ Üniversitesi Doktora Tezi, 2023, p. 101.

²¹¹ Engin Yıldırım et al., “Makine Öğrenmesi Yoluyla Sendikal Haklarla İlgili Olarak Anayasa Mahkemesine Yapılan Bireysel Başvuruların Analizi” *Anayasa Hukuku Dergisi*, Cilt 10, Sayı 20, 2021, p.479.

²¹² Mumcuoğlu et al., p. 12.

²¹³ Micheal Benedict L. Virtucio et al., “Predicting Decisions of the Philippine Supreme Court Using Natural Language Processing and Machine Learning”, *2018 42nd IEEE International Conference on Computer Software & Applications*, 2018, p. 134.

²¹⁴ Ruger et al., p. 1152.

²¹⁵ Virtucio et al., p. 134.

B. SUPPORT VECTOR MACHINES

Support vector machines are put forward as a type of learning machine.²¹⁶ They are described as mapping vectors into some high dimensional feature space and creating a linear decision surface that ensures high generalization.²¹⁷ This means that, fundamentally, a support vector machine is a method by which the process of splitting points in a space, or classification of them, is handled.

Octavia Maria Şulea and others utilised SVMs in predicting French Supreme Court cases, where they dealt with a non-binary classification; they have run their prediction model on a 6-class and 8-class basis separately.²¹⁸ They identified their labels for these classes by eliminating all labels which had less than 200 examples.²¹⁹ Their features were word occurrences (unigrams and bigrams) coupled with other features such as the ratio of how many unique words there are to total number of words.²²⁰ Using these, they have managed a 96.9% accuracy rating in predicting the result of a French Supreme Court case.²²¹ They cite the lack of interpretive freedom in French legal system that allowed the courts to be more predictable as a reason for their high accuracy.²²²

In “Predicting judicial decisions of the European Court of Human Rights: a Natural Language Processing perspective” by **Nikolaos Aletras and others**, the problem becomes a binary classification aiming to predict violation or non-violation by the

²¹⁶ Vladimir Vapnik and Corinna Cortes, “Support-Vector Networks”, *Machine Learning*, Vol. 20, 1995, p. 274.

²¹⁷ Vapnik and Cortes, p. 274.

²¹⁸ Octavia-Maria Şulea et al., “Predicting the Law Area and Decisions of French Supreme Court Cases”, arXiv:1708.01681v1, 2017, p. 718.

²¹⁹ Sulea et al., p. 718.

²²⁰ Sulea et al., p. 719-720.

²²¹ Sulea et al., p. 720.

²²² Sulea et al., p. 720.

ECtHR.²²³ They have modelled their features as word sequences and topics,²²⁴ citing specifically Bag-of-Words model where a document is considered simply as a collection of words without order,²²⁵ the same approach used by **Şulea et al.** study.²²⁶ They manage to achieve a 79% accuracy on average.²²⁷

Virtucio et al., aiming a binary classification, had achieved a 55% accuracy,²²⁸ even though they were only aiming to predict either of two scenarios, akin to **Aletras et al.**; that is affirmed or reversed. **Görentaş** manages to achieves %81 accuracy²²⁹, **Yıldırım et al.** manages %63²³⁰ whereas **Mumcuoğlu et al.** manages around 73% balanced accuracy, with the highest predictive power in cases belonging to Court of Appeal on Taxation: 90%.²³¹

SVM approaches manage relatively high accuracies, highlighting a close connection of legal decisions with the words but not necessarily meaning. In both **Şulea et al.** and **Aletras et al.**, the data is considered only as a collection of words instead of a syntactically sound structure with any semantics attached to it. The question of if some sequences of words have appeared or not seemingly enables for the outcome of a case to be predicted. The variance of the results may be due to many factors such as differences in legal systems, how cases summaries are written, the type of cases in question and so forth.

²²³ Nikolaos Aletras et al., “Predicting judicial decisions of the European Court of Human Rights: a Natural Language Processing perspective”, *PeerJ Comput. Sci.*, 2:e93, 2016, p. 9.

²²⁴ Aletras et al., p. 8.

²²⁵ Aletras et al., p. 8.

²²⁶ Sulea et al., p. 719.

²²⁷ Aletras et al., p. 2.

²²⁸ Virtucio et al., p. 133.

²²⁹ Görentaş, p. 101.

²³⁰ Yıldırım et al., p. 479.

²³¹ Mumcuoğlu et al., p. 12.

C. DEEP LEARNING

Deep learning dates back to 1940s as a technology, even though it had been known by different names such as cybernetics or connectionism.²³² Deep learning can be expressed as an approach to AI, as “technique that allows computer systems to improve with experience and data”, where world is represented as a “hierarchy of concepts”, “with each concept defined in relation to simpler concepts, and more abstract representations computed in terms of less abstract ones”.²³³ In general, deep learning has a visible layer for its input in which the data exists as it is, then hidden layers where features of data are extracted, and finally an output layer in which the data is classified.²³⁴

Deep learning utilises layered neural networks, and a type of these networks is recurrent neural networks, specialised to be used in processing of sequential data.²³⁵ As such, recurrent neural networks are well adapted to dealing with text, as any text can be expressed as a sequence of characters, syllables, words, and so forth. Neural networks are a network of neurons where each neuron holds a value and in a simple feed forward neural network this value is propagated from the input layer, through the hidden layer, and finally to the output layer.²³⁶ Recurrent neural networks differ from these as they have recurrent feeds to the current state from previous states,²³⁷ which allows them to have a memory.

There are three variants of deep learning approaches utilised in the scope of this survey for legal prediction and these are LSTM, BiLSTM, and GRU. The memory of a recurrent neural network is limited, and long-short term memory version of RNNs are

²³² Ian Goodfellow et al., *Deep Learning*, MIT Press, 2016, p. 13.

²³³ Goodfellow et al., p. 8.

²³⁴ Goodfellow et al., p. 6.

²³⁵ Goodfellow et al., p. 373.

²³⁶ Ralf C. Staudemeyer and Eric Rothstein Morris, “Understanding LSTM – a tutorial into Long Short-Term Memory Recurrent Neural Networks”, arXiv:1909.09586v1, 2019, p. 2.

²³⁷ Staudemeyer and Morris, p. 2.

proposed as a remedy for this problem.²³⁸ Another variant of this gated version of RNNs is gated recurrent units proposed by **Cho et al.** in 2014.²³⁹ However, despite their differences, all these systems are trained, fundamentally, by encoding the data, predicting the next item in the sequence, and then by comparing their outputs to actual values and adjusting accordingly.²⁴⁰²⁴¹ Since all three methods are but variants of the RNN approach they will be examined together.²⁴²

For deep learning, **Mumcuoğlu et al.** leveraged the word embeddings method which assumes that “the meaning of a word, to some degree, can be inferred from its statistical co-occurrence with other words”.²⁴³ This methodology achieves differing results for each algorithm adopted; ranging from 68% to 73%.²⁴⁴

Shangbang Long and others and **Kankawin Kowsrihawatt and others** take a different approach to the problem of prediction than the above studies and both include legal rule text in their models.²⁴⁵²⁴⁶ **Long et al.** approaches the problem with considering

²³⁸ Sepp Hochreiter and Jürgen Schmidhuber, “Long Short-Term Memory”, *Neural Computation*, Vol. 9, Issue 8, 15 Nov. 1997, p. 1.

²³⁹ Junyoung Chung et al., “Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling”, arXiv:1412.3555v1, 2014, p. 1.

²⁴⁰ Alex Graves, *Supervised Sequence Labelling with Recurrent Neural Networks*, *Studies in Computational Intelligence*, vol. 385, Springer, 2012, p. 19-20, 29.

²⁴¹ We have to note that this is an extremely simplified explanation derived to the best of our abilities from Graves (see *supra* note).

²⁴² There are also convolutional neural networks, which operate differently than RNNs and have found place in some studies (e.g. Long et al. and Lage-Freitas et al.). While we have discussed Long et al. we left Lage-Freitas et al. out of this study as it was similar to most other work we have surveyed, along with convolutional neural network architecture as it has limited usage for this specific field of application (see our explanations for RNNs) and as they also are a method of deep learning they are not completely different than RNNs.

²⁴³ Mumcuoğlu et al., p. 9.

²⁴⁴ Mumcuoğlu et al., p. 12.

²⁴⁵ Shangbang Long et al., “Automatic Judgement Prediction via Legal Reading Comprehension”, arXiv:180906537v1, 2018, p. 558.

²⁴⁶ Kankawin Kowsrihawatt et al., “Predicting Judicial Decisions of Criminal Cases from Thai Supreme Court Using Bi-directional GRU with Attention Mechanism”, *2018 5th Asian Conference on Defense Technology (ACDT)*, 2018, p. 50.

the inputs as a trio: the facts, the plea, and the law.²⁴⁷ Their algorithm considers all of these and then predicts based on the interaction between these.²⁴⁸ On the other hand, **Kowsrihawwat et al.** applies the GRU model to law text and fact text separately, calculate attention scores for each of them which represents their relevance and then this score is multiplied with them before they are combined, which they claim imitates a judge's legal interpretation.²⁴⁹ **Long et al.** has achieved around 82% accuracy with their own deep learning approach in predicting civil cases of Supreme People's Court of People's Republic of China,²⁵⁰ and **Kowsrihawwat et al.** managed to achieve roughly 66% accuracy in predicting criminal cases of Thai Supreme Court.²⁵¹

In all of these cases, the decision making is not syllogistic. Even though **Long et al.** and **Kowsrihawwat et al.** bring legal rules into the mix, they do not apply syllogism in any part of their procedure, instead they add the rule text to their statistics-based approach. **Mumcuoğlu et al.** on the other hand does not even incorporate the rule text separately. Still, these studies manage to predict the case outcomes with their solely statistical methodologies.

II. PATTERN RECOGNITION AND PROBABILITIES

The three methods we have investigated —that is; DT, SVM, and DL— all leveraged the fact that there is some connection between what they examined and what they aimed to predict. However, the connections they sought were never based on how a

²⁴⁷ Long et al., p. 559.

²⁴⁸ Long et al., p. 562.

²⁴⁹ Long et al., p. 51.

²⁵⁰ Long et al., p. 567.

²⁵¹ Long et al., p. 54.

law would be applied to a concrete case. Instead, they have managed to find seemingly arbitrary connections.

In the case of **Ruger et al.** the interpretability of decision trees made it transparent that there is a connection between the circuit of origin and the decision, with very little other factors contributing to a better prediction than the legal experts. Considering their apparent connections between that led to predictions, a legal interpretation seems almost secondary in nature to the decisions.

In the case of support vector machines, the connection between the words related to the case seems ultimately connected to the decision. These connections that enabled the predictions, did not consider any syntax, but operated by merely considering the occurrence of words. The machine managed to classify the word occurrences common to cases in certain groups accurately without doing any rule-based analysis.

In the case of deep learning, and all associated methodologies, even when rule text was involved in the model, the machine did not make any logical questioning. Meaning of words were associated with each other through their co-occurrence and using that and a lot of experience in the form of how human judges decided cases before gained in training, machine managed to make accurate predictions.

All three models utilise varying data ranging from the judges deciding the cases to the laws of the jurisdiction that would be applied to the case. Most of them especially focus on analysis of word use in decisions and fact summaries. This leads to an orientation that is heavily influenced by words but not language in general. During the analysis of the training data, there have been processing of the words included in the data such as removing fixed words, converting all characters to lowercase, removing of words and

characters without meaning, converting words into stems, and so forth.²⁵²²⁵³ After the utilisation of these methods, researchers focus on converting word relations into matrices.²⁵⁴²⁵⁵ This makes it so that the text is almost devoid of any meaning in a traditional linguistic approach.

It must be noted that while these models also seem decision-centric in their application, and as such more suited for common law applications, they have been used in civil law settings as well. **Görentaş, Yıldırım et al.**, and **Mumcuoğlu et al.** are examples to from varying Turkish high courts studying Court of Jurisdictional Disputes, synodical rights disputes decided by the Supreme Court, and various high courts ranging from first instance appellate courts for civil law, criminal law, administrative law cases and Supreme Court decisions respectively. **Virtucio et al.** has worked on the Philippine legal system which is a mix between common law and civil law, **Şulea et al.** works on decisions in French legal system which is a civil law system such as **Kowsrihawatt et al.** with Thai legal system and **Long et al.** with Chinese legal system.

The models are not decision-centric per se because of two reasons: (1) the models are sometimes fed legal rules as well and as such are able to “infer” connections from rules, and (2) the models are often fed extra-legal factors as well, which would not be a consideration in a decision-centric approach. In analysing precedents, the precedent is considered for legal opinions. On the contrary, AI does not have any understanding as to the legal matters of the case—to be more direct, AI does not have any understanding as to anything in the traditional sense to our knowledge.

²⁵² Görentaş, p. 92.

²⁵³ Yıldırım et al, p. 478.

²⁵⁴ Görentaş, p. 92-93.

²⁵⁵ Yıldırım et al, p. 478.

All of the models we have discussed work by the premise of recognising certain patterns in decisions. They are inherently probabilistic in the sense that they infer patterns and assume a decision by using the similarities in the current data to past data. These models are still in progress and extremely unreliable yet provide results that might lead to their adoption in everyday use. In the following chapter we discuss previous discussions about these models and how they relate to legal theory, and construct our own arguments through what we have noted about AI models in this chapter and what we have noted about Realism in the previous chapter.



CHAPTER 3:

AI FROM THE LENS OF LEGAL REALISM

In this chapter we deal with the question of how Legal Realism would deal with the application of AI in legal decision making and prediction. In previous chapters we have laid out the core tenets of Legal Realism and tried to summarise how AI is currently being used in legal decision prediction. Here, we are going to try and bring these together and create a discussion of whether Legal Realism can provide a theoretical framework for the relevant AI applications.

In the first section, we are going to discuss some thoughts in the literature that point to the connection between Realism and AI. The so-called “rule-nihilism” of Realism make it so that it is the first candidate that comes to mind when we consider how AI operates. However, there is still a gap in the literature regarding this issue despite there being discussions of AI going back to more than half a century ago.²⁵⁶

In the next section we are going to consider if Legal Realism is compatible with AI. To that end, we will first discuss where AI lacks in a legal decision making setting. After which, we will try and explain how our understanding of Legal Realism can be applied to AI models and why we consider Legal Realist theory fit for this purpose.

I. FIRST RESPONDERS

A. LEGAL DETERMINISM

One of the most comprehensive discussions on the subject of legal theory and AI can be attributed to **Maxi Scherer**. In “Artificial Intelligence and Legal Decision making: The Wide Open? Study on the Example of International Arbitration”, **Scherer**

²⁵⁶ See Lawlor.

discusses, among other issues, Legal Realism and legal formalism in the light of decision making with AI. While **Scherer**'s is not the first to discuss the issue at hand, we will first try and break down what this paper argued, as it is arguably the closest one to our study.

Scherer's paper focuses on our issue under a part titled "Paradigm-Shift in Legal Decision Making: Probabilistic Inference Instead of Deductive Reasoning and Logic".²⁵⁷ This title is noted for the use of "paradigm-shift", which we will discuss as well. We will ask what is implied by "scientific paradigm" and whether Law can be subject to such a paradigm. In a sense, relating this question to our main question of Legal Realism as a framework for understanding.

Scherer begins with citing **Reed C. Lawlor**, who was one of the first people that speculated the involvement of computers in legal affairs.²⁵⁸ **Scherer** notes the reservation by **Lawlor**, that is "reliable prediction would depend on a 'scientific' understanding of the ways the law and the facts impact on the judges' decision."²⁵⁹ That is, however, a misrepresentation of **Lawlor**'s argument. **Lawlor** argues that "The ultimate successful use of computers in prediction of decision depends on the extent to which scientific methods are applicable to law."²⁶⁰ This does not necessarily imply "reliability" per se but steers the discussion towards the analysis of Law as a science. **Scherer**, after noting this reservation by **Lawlor**, notes that "Even today such 'scientific' understanding of judicial decision making lacks and is a debated topic amongst legal philosophers and

²⁵⁷ Maxi Scherer, "Artificial Intelligence and Legal Decision making: The Wide Open? Study on the Example of International Arbitration", *Journal of International Arbitration*, 2019, p. 25.

²⁵⁸ Lawlor, p. 337.

²⁵⁹ Scherer, p. 25.

²⁶⁰ Lawlor, p. 338.

theorists.”²⁶¹ On that, while the scientific nature of Law is out of the scope of this thesis, this question will be reiterated further in this chapter.

Scherer, then splits the legal theory into two categories: those that argue for the use of logic for the interpretation of preexisting rules, legal formalism; and those that emphasize the importance of extra-legal factors and the political dimension of the law, Legal Realism.²⁶² **Scherer** posits this distinction in such a way that all legal theories are either in one group or the other. However, grouping theories such as legal positivism or natural law under these two broad categories would be oversimplification. Legal positivists such as **H.L.A. Hart** are known to criticize both points of view in detail.

Following this, **Scherer** argues that “the use of AI in legal decision-making does not fit easily in either category.”²⁶³ Other legal theories do not fit easily in either category, and overlooking this while arguing neither Realism nor Formalism can explain the use of AI in legal decision making would be misleading. **Scherer’s** distinction oversimplifies, or in a way overcomplicates Legal Realism as well. We have tried to establish what Legal Realism entails, and while it is true that Legal Realism emphasizes the importance of extra-legal factors; it also asks for a pragmatic approach to Law and calls for measures in prediction of legal decisions.

Following with laying out formalism, **Scherer** summarizes legal formalism as an outlook that considers law as a self-contained system, both in what is law and what law should be.²⁶⁴ Legal formalist outlook will consider a judicial decision to be “the product of a seemingly mechanical or mathematical application” of rules.²⁶⁵ The formula

²⁶¹ Scherer, p. 25.

²⁶² Scherer, p. 25.

²⁶³ Scherer, p. 25.

²⁶⁴ Scherer, p. 25.

²⁶⁵ Scherer, p. 25.

is considered to be that the decision is the application of rules to the formula, or in legal syllogism the premise of a fact occurring and that fact being part of a preexisting rule which stipulates an outcome if the fact has occurred.²⁶⁶ **Scherer** cites **Hart** and the distinction **Hart** made between hard cases and clear cases as an example of influence of formalism on **Hart**, for the case of clear cases.²⁶⁷

Scherer argues that AI processes would “run counter to this understanding of legal decision-making”.²⁶⁸ **Scherer** says that while some models are rule-based and apply logic and deductive reasoning, the others avoid these methodologies.²⁶⁹ We must note, the models **Scherer** refers to as using rule-based approaches are not used for legal decision making, and cannot be used for such purpose, at least on their own, for reasons we have discussed previously, such as complexity. That aside, **Scherer** correctly points out that models in use (citing machine learning and neural networks) have no pre-defined rules, at least in the legal sense.²⁷⁰ **Scherer** puts forward two ideas: (1) “machine learning program extracts the algorithm from the observable data” as “deductive, causal reasoning is [...] replaced by an inverse approach”, and (2) “rather than using logic, the AI model calculates probabilities”.²⁷¹

The two ideas **Scherer** opens to discussion are seemingly contradictory: if an AI system extracts an algorithm from data that allows it to decide on a case, how is it so that we can claim it does not use logic; for an algorithm refers to a “step-by-step procedures for solving a problem”; this is what legal formalists argue that must happen in the first place, a set of procedures must apply to determine if a rule applies to a case. The

²⁶⁶ Scherer, p. 25.

²⁶⁷ Scherer, p. 25-26.

²⁶⁸ Scherer, p. 26.

²⁶⁹ Scherer, p. 26.

²⁷⁰ Scherer, p. 26.

²⁷¹ Scherer, p. 26.

contradiction arises because **Scherer** seems to think that calculating probabilities does not employ logic. On the other hand, **Scherer** must be challenged in proposing that “causal reasoning is replaced by an inverse approach”.²⁷² It is true that AI adopts an algorithm, however that algorithm cannot be said to be extracted, and causal reasoning is not inverted, it is replaced by an entirely new approach.

Accepting for a moment, that the normal approach is that an algorithm is applied to a case to reach to an outcome, the inversion of that means that the same algorithm is *extracted* from the collection of given inputs and outcomes. AI models, as we have laid them out, do not do this, they do not output an algorithm. What they do is that they come up with an algorithm that is created for an entirely different situation than what the legal formalists have envisioned. So, in agreeing with **Scherer**, our opinion is concurring.

Scherer ultimately argues the following on legal formalism: (1) “applying such machine learning processes in the legal decision making context [means] a departure from [...] formalist theories”, (2) “a decision based on those AI models would not be based on pre-determined legal rules”, (3) “[it] would not be the result of deductive logic”, and (4) it would not adhere by legal syllogism”.²⁷³ A final note to this result, except from our previously discussed deviations in the basis of the opinions, is that “a decision based on those AI models” *could* be based on “pre-determined legal rules”. It is just so that (1) our insight towards some of the AI models is limited, (2) to the best of our knowledge AI models are not *directly* based on those rules. We have to accept, however, as is the case with judges, AI is influenced by the rules, as they inevitably influence its dataset.

²⁷² Scherer, p. 26.

²⁷³ Scherer, p. 26.

On Legal Realism, **Scherer** draws attention to how legal formalism was susceptible to criticism and the role of Legal Realism in fulfilling that role.²⁷⁴ **Scherer** accurately points out that Realist theories varied significantly, which is an important point to consider when examining any issue under a Realist lens. After that, the description of Realism is made as follows: “[attack on] the idea that the law was a mechanical application of pre-determined rules by the judge by means of logic”, “[the idea that] rules do not play a determinative part in legal decision-making”, and “[the argument] that logic and rule-deference is only a façade and ignores social interests involved”.²⁷⁵

Scherer follows these tenets of Realism with the account of **Holmes**, with an emphasis on the emphasis of **Holmes** on prediction.²⁷⁶ **Holmes** is to be considered a key figure for Realist thought, a progenitor. We have previously touched on the prediction emphasis of **Holmes** as well. **Scherer**’s following thoughts almost completely match the arguments of this thesis:

Holmes’ emphasis in 1897 on prediction and statistics in legal decision-making, in lieu of logic, shines today in new light when considered the implications of AI. As discussed above, prediction based on statistics or probabilities are precisely features used in AI machine learning models. Moreover, the importance of extraneous non-legal factors, as argues by the legal realists, is confirmed by the predictive AI studies, cited above. In the ECtHR study, the part of the judgements with the highest predictive value is not the legal section but the section relating to the factual background. Also, the US Supreme Court study

²⁷⁴ Scherer, p. 26.

²⁷⁵ Scherer, p. 27.

²⁷⁶ Scherer, p. 27.

included in the computer model extra-legal elements such as the judges' political preferences.²⁷⁷

However, a divide follows after this summarisation. For **Scherer**, “this goes well beyond legal realist theories”, and AI thus does not vindicate Legal Realism, even when considering what AI is doing is what human judges have always done; as these, ignore the central place of probabilities as a normative basis.²⁷⁸

Another distinction of extreme importance is taken into account by **Scherer**, that legal theories have separate descriptive and normative aspects.²⁷⁹ The separation between “is” and “ought” in legal theory is central to much debate, and **Llewellyn** has pointed out that is-ought distinction is also central to Legal Realism.²⁸⁰ **Scherer** notes this separation is as it relates to “how judges *do* effectively reason and take decisions” and “how they *should* reason and take decisions”.²⁸¹

Formalism, as **Scherer** argues, includes both a descriptive and normative approach to the subject, that is to say that the formalist argument is that “judges apply the law as a matter of logic, deduction and legal syllogism” in fact, and that is also how it should be so that the law is not impacted by extra-legal factors such as politics.²⁸² As opposed to that, according to **Scherer**, Legal Realism only entails a description of how judges decide.²⁸³ In **Scherer**'s words:

[they] trace the *reality* of judicial decision-making — hence the name of the movement [...] they do not go as far as arguing that judges *should* take into

²⁷⁷ Scherer, p. 28.

²⁷⁸ Scherer, p. 28.

²⁷⁹ Scherer, p. 28.

²⁸⁰ Llewellyn, 1931, p. 1231.

²⁸¹ Scherer, p. 28.

²⁸² Scherer, p. 28.

²⁸³ Scherer, p. 28.

account extraneous non-legal factors [...] while it might be a matter of fact judges are influenced by extraneous factors, such as food breaks, no one seriously argues that this is a good thing and should be the normative basis for judicial activity.²⁸⁴

This is not exactly true. While it is true that no one seriously argues that food breaks should be a normative basis for legal judgments; Realists such as **Hutcheson**, who were also practicing justices, were proponents of use of “dice” in legal decision making, an example account being the following account:

And not only do I set down boldly that I, “even as your other worships do,” invoke and employ hunches in decisions, but I do affirm, and will presently show, that it is that tiptoe faculty of the mind which makes [...] the best lawyers [and] which have widened and will ever widen man’s horizon.²⁸⁵

So, it would not be the entire truth to say Realists do not seriously argue that usage of extra-legal factors is not a “good thing”; and even considering that if they did not argue that this is a “good thing” per se, the way **Scherer** puts it would mean the Realist perspective is supporting the opposite, that this is a “bad thing”, which would be normative as well.

We have gone to great lengths to establish that there is no monolith school of Realism. **Llewellyn** made clear that Realists give importance to the fact that Realism is concerned with how judges decide, and not how *they should* decide. So, it would be contradictory to lean on **Hutcheson** and claim that Realists argue normatively that this is how judges should be deciding. However, we have also made it clear that Realism has a

²⁸⁴ Scherer, p. 28.

²⁸⁵ Hutcheson, p. 279.

normative aspect regarding how jurists or lawyers approach Law, in the sense that their task is a task of prediction; and they have a normative agenda regarding the study of the law. With that, we have also established that the task of legal decision making or legal decision prediction; the task of both the judge and the lawyer and the jurist alike is a complex task which cannot be delegated to faculties of simple calculations. We must argue then that while Legal Realism does not have a prescriptive approach to how judges decide, a “realist” approach to law reveals that in accepting the Realist reality, there is no normativity to be discussed. This is how judges decide because this is how they are able to decide.

A consideration of how judges decide cases, or how they can decide cases is closely connected to **Hart**’s arguments regarding “hard cases” and “clear cases”. **Hart** does argue that in some cases law is incomplete, and in those cases, judges must employ their discretion, yet this is only possible in cases where the open-texture nature of language and law requires it.²⁸⁶ Realist perspective does not require such distinction as they argue that the law is not certain and thus creating an arbitrary area where the open-texture nature of language requires discretion is redundant. **Allan C. Hutchinson** reminds us that “[Hart] admits that it is a matter of considerable difficulty to provide an exhaustive account of what makes a ‘clear case’ clear and what makes a ‘hard case’ hard.”²⁸⁷

To claim Legal Realism has a normative position on how cases should be decided is not the goal of our argumentation. Rather, we would argue that Legal Realism does not need a normative approach to how cases should be decided. Furthermore, this leads to a larger question, which will be out of scope of this thesis, on the topic of if scientific

²⁸⁶ H.L.A. Hart, *The Concept of Law*, Oxford: The Clarendon Press, 2nd ed., 1994, p. 252.

²⁸⁷ Allan C. Hutchinson, “A Postmodern’s Hart: Taking Rules Sceptically”, *The Modern Law Review*, Vol. 58, No. 6, 1995, p. 793.

frameworks need normative positions on the subject of their studies. There is another question that is relevant on this topic, which is if Law has a scientific aspect to it, which created and still demands much discussion, on which we will humbly propose an idea at our conclusion.

Disregarding the argument regarding the requirement of a normative aspect for Legal Realism, **Scherer**'s next argument regarding the normative aspect of AI models is also up for debate. **Scherer** argues that, considering the conclusions we have summarised above under the discussion against legal formalism, "[the conclusions] would be true on a descriptive level and on a normative level"; that is "how these models do effectively decide" and "how these models should decide".²⁸⁸ **Scherer** then suggests that replacing the normative basis of judicial decision making "goes well beyond legal realists' theories".²⁸⁹

It is not clear how **Scherer** comes to this conclusion, the question of why this would be true on a normative level is never answered. Why is it so that, given the AI models of today, how they decide should be the normative basis of how legal cases should be decided? One is to assume that because "there is a need for normative aspect" and in using AI as decision making machines, we would have adopted their way as the way this task *should* be done. This is neither true nor necessary.

After claiming that "probabilities" will be the normative basis for AI based legal decision making, **Scherer** then argues about the problems of this approach. According to **Scherer** a probabilistic approach entails problems such as "AI models [taking] conservative approaches" or "probabilities or statistics [not being] an accepted legal basis

²⁸⁸ Scherer, p. 29.

²⁸⁹ Scherer, p. 29.

for decisions’’.²⁹⁰ These hold true, but they only suggest further normative problems. They are related to as how cases should or should not be decided and must be severed from the body of our arguments by Hume’s Guillotine.

Aside from these, there are some issues we would highlight in **Scherer’s** arguments that we think relate to *technical* issues rather than normative. For example, **Scherer** argues that AI based decisions lack reasoning.²⁹¹ This is true in the current level of AI based predictions, however, that does not mean in a scenario where we employ AI for the use of legal decision making, we cannot couple it with a reasoning machine, which would, as **Hutcheson** accounts,²⁹² ex post facto create rationale for decisions using logomachy. **Scherer** also points out problems such as “what threshold would be appropriate above which a claim is deemed granted” or the building up of bias against parties.²⁹³ These are real issues in AI based legal decision making, but these are issues which technical improvements upon AI models can solve. The question in the legal theory sense is there to ask, “how this decision come to be”, which will be answered as “building up of bias”, whereas the question regarding “what can we do about bias” must be answered by AI experts in the form of “improve our training algorithms to account for bias”.

At the end **Scherer** concludes that accepting probabilities as the normative basis for legal decision making will be a paradigm-shift in legal theory, which **Scherer** calls legal determinism since “it determines future outcome on probabilistic calculations based on past data”. Disagreeing with **Scherer** on the normative basis aspect, from the Legal Realist perspective we have established thus far, we would argue that deciding on “future

²⁹⁰ Scherer, p. 30.

²⁹¹ Scherer, p. 29.

²⁹² Hutcheson, p. 280.

²⁹³ Scherer, p. 30.

outcome on probabilistic calculations based on past data” is well in the scope of Realistic perspectives. Realists have touched on the impact of precedents on legal decisions before and criticised it as well.

B. A CLOSER INSPECTION

Ruger et al. is one of the first modern studies that deal with the application of artificial intelligence techniques in legal decision making. We have summarised their findings above. With that, this section will focus more on their theoretical discussions, which begin with **Holmes**.²⁹⁴ They cite to the Holmesian “prediction theory of law”; which defined the law as the “prophesies of what the courts will do”.²⁹⁵

As we have previously discussed, **Holmes** was a progenitor for Legal Realism, and they follow with this as well: “Holmes and his followers undermined the classical notion of law as a set of static, natural, and apolitical rules that could be mechanically discerned and applied by judges”.²⁹⁶ They build upon **Holmes** and emphasise the limitations of Holmesian theory, which lack practical details and only proposes conventional methods.²⁹⁷ They argue Realist theory suffers “in terms of positive predictive theory”.²⁹⁸

They are correct to point out that Legal Realism suffers in majority from ingenuity in pointing out exactly how judges arrive at decisions, except from accounts directly coming from judges. Still, the lack of empirical research by legal theorists does not mean that Legal Realism is not suited to provide a framework for any empirical

²⁹⁴ Ruger et al., p. 1155.

²⁹⁵ Ruger et al., p. 1155.

²⁹⁶ Ruger et al., p. 1155.

²⁹⁷ Ruger et al., p. 1156.

²⁹⁸ Ruger et al., p. 1156.

research. On the contrary, it is because Realism promotes such analysis, Legal Realists are seen lacking in this regard and are criticised in not following **Holmes**.²⁹⁹

Another study on legal decision making with AI that discusses the theoretical aspect of the endeavour is **Aletras et al.**, in which they open with the result that their results strongly imply the correlation between facts of a case and the result is stronger than the correlation between legal reasons and the result.³⁰⁰ Following this, they delve into the discussion of Legal Formalism and Realism, in saying that:

These results could be understood as providing some evidence for judicial decision-making approaches according to which judges are primarily responsive to non-legal rather than to legal, reasons when they decide appellate cases.³⁰¹

They note that the debate between legal formalism and Legal Realism is out of scope of their paper.³⁰² After which they provide basic understandings of Formalism and Realism, describing Realism as the view of judges “responding to the stimulus of the facts of the case”.³⁰³ This particular view of Realism is simple enough to not warrant any further explanation, but it requires emphasis as it simplifies what we have said until now.

Aletras et al. points out that previous research has empirically showed that “pure legal models, especially deductive ones” are falsified, for “courts further up the hierarchy”.³⁰⁴ **Aletras et al.** mentions that it has been previously discussed that “best way to explain past decisions” is by emphasizing non legal factors that impact judges; and

²⁹⁹ Ruger et al., p. 1156.

³⁰⁰ Aletras et al., p. 11.

³⁰¹ Aletras et al., p. 11.

³⁰² Aletras et al., p. 11.

³⁰³ Aletras et al., p. 11.

³⁰⁴ Aletras et al., p. 12.

cites **Llewellyn** as an example of Realists trying to classify cases by other variables “that did not reflect standard legal doctrine”.³⁰⁵

Aletras et al. manages to put forward a reasonable understanding of Legal Realism, and then manages to connect that idea to previous data. While their work is not focused on the theoretical aspect, they propose that their “results could be understood as lending some support to the basic Legal Realist intuition according to which judges are primarily responsive to non-legal, rather than to legal, reasons when they decide hard cases.”³⁰⁶ We have to note that this is not our argument (which we discuss in detail below), while we see this as a compatibility regarding Legal Realist application to legal decision making with AI. **Aletras et al.** also admits that they have merely presented a “simplified snapshot of a very complex debate”³⁰⁷,

In the end, **Aletras et al.** suggests that “not more should be read into this than [their] results allow” and try and make reservations as to their results.³⁰⁸ They warn about the “selection effect” which “pertains to cases judged by the ECtHR as an international court”.³⁰⁹ They warn that it could be so that since these cases are filtered through many layers coming from the national level to the Chamber, judges could be focusing more on the facts as the legal rules could have already been considered as “indeterminate”.³¹⁰ They note “further text-based analysis is needed in order to determine whether the results could generalize to other courts”.³¹¹ We know, as we have studied other work, that there is no issue regarding a selection bias as works conducted by many others point to the same

³⁰⁵ Aletras et al., p. 12.

³⁰⁶ Aletras et al., p. 12.

³⁰⁷ Aletras et al., p. 12.

³⁰⁸ Aletras et al., p. 12.

³⁰⁹ Aletras et al., p. 12.

³¹⁰ Aletras et al., p. 12.

³¹¹ Aletras et al., p. 12.

truth as theirs did. There are other factors in work other than those “purely legal”, and in order to explain these, there is a need for a framework in the form of Legal Realism.

Zaky approaches the matter in the opposite direction, arguing that while Realism at first glance is seemingly correlated with AI applications, Realists would face problems in a theoretical approach:

The existence of the realist empirical project and Holmes’ emphasis on the role of predication in law makes it easy to draw connections with artificial intelligence “a legal duty so called is nothing but a prediction that if a man does or omits certain things, he will be made to suffer in this or that way by judgment of the court; and so of a legal right.” Realism is inherently a reactionary theory that is concerned with the functional aspect of the law i.e. how the law is and not what it should be. Accordingly, realists would have a hard time assessing the theoretical implications of AI and would most probably divert their attention to the tangible AI-run examples: US Supreme Court, ECtHR and COMPAS.³¹²

We agree with **Zaky** in the sense that it is indeed true Realists would divert their attention to the tangible AI-run examples, and it is what we are doing in this study. However, there is absolutely no reason for theory to not engage with tangible examples.

Zaky also discusses the similarities between the Realist approach and AI naming first the empirical study of law, then the pressure of facts in decisions as opposed to laws in both AI and Realist theory, and the inability to track down the sources of hunch in both Legal Realist analysis of judges and AI research.³¹³ We believe that this approach is *correct* to some degree but not *fruitful*. While the results in AI research may correlate with

³¹² Zaky, p. 53.

³¹³ Zaky, p. 53.

the Realist ideas, as **Aletras et al.** also noted these correlations are (a) anachronistic and (b) not definitive empirical results. AI project has not proven the Realist theses, nor does it seem to aim to do that. **Zaky** also notes that the AI models do not prove the accuracy of the Realist approach as *artificial intelligence* differs from our own *intelligence*.³¹⁴ The similarity exists merely at surface.³¹⁵

At the end, **Zaky** falls in a completely opposite perspective to our thesis, claiming the following:

Ultimately, exploring AI within realism, to somehow substantiate or dismiss realism's stature in legal theory is nonsensical, as AI's potential far exceeds realism's perception of the law. This is not a critique within any stretch of the imagination. To the contrary, AI's potential is not a concept realism can grapple with because, as a theory, realism is mainly concerned with how the law exists today. Accordingly, realism can only be explored by resorting to the examples of the US Supreme Court, ECtHR, COMPAS and companies such as ArbiLex because these are the practicing examples of AI.³¹⁶

The issue is that **Zaky** does not seem to give any plausible answer as to why exploration of AI does not work in substantiating or dismissing Realist claims. His approach is of that AI's potential exceeds the Realist perception and since Realism is concerned with law as it exists today it cannot understand it ever. This is entirely tautological as it seems to argue that if a theory is concerned with *now* it can never understand *now* as *now* will always be *past* in the *future*. We will argue that the opposite is true, since Realism is interested in

³¹⁴ Zaky, p. 54.

³¹⁵ Zaky, p. 29.

³¹⁶ Zaky, p. 55.

now, it is best fit to understand these tools as they are and as they can be, as it is not stuck in a predetermined understanding and adapts to life and law in motion, in flux.

Zaky also notes how AI systems differ from the Realist claims.³¹⁷ Yet, they differ naturally. Realists investigated the law as it was then, yet things are of course different now. As we have tried to point out, Realism should not be understood as what they have found out about law but what questions they have asked about the law.

The contradiction of our argument with **Zaky** seems to exist due to our agreement in that how AI models work should not be used as an example of Realist understanding of law and to substantiate its thesis as they regard to how judiciary works in a retrospective manner. We believe that “exploring AI within realism” will substantiate the Realist claims not because Realist claims are correlating with what we have seen but because we believe Realist exploration of AI will provide new insights as to how these models work in a legal theory perspective. As such, the parallels between the Realist claims and AI³¹⁸, to our purpose is not valued in a retrospective but as a prospect into understanding of a new way of how courts decide, in fact.

II. A COMPATIBLE COUPLE

We have discussed above how others have remarked on Realism as a theoretical basis for legal decision making with AI. In arguing with them, we have proposed much of our own ideas as well. However, in our arguments we have focused mostly on their explanations for Legal Realism, and much less on specifics of the case at hand, how AI decides (or predicts) cases. In the following section, we will focus on discussing the idea

³¹⁷ Zaky, p. 55.

³¹⁸ Zaky, p. 55.

that *if* legal decision making with AI were the norm, then we would be able to best explain the processes that govern it using Legal Realism.

These models, as they are, are not deciding on judicial decisions, they are predicting them. However, since their predictions can emulate that of a human judge's decisions, there is a possibility that these models will be eventually used for legal decision making. A legal decision is how the judiciary authority is realised.³¹⁹ To our purpose, the legal decision refers to the decision of a court on a judicial dispute, the final verdict.

We must note, this thesis does not suggest AI as an experimental application of Legal Realist ideas, or their redemption, neither does it suggest that legal decisions made by AI are proper legal decisions simply because they are able to emulate human judges. This must be true for human judges as well, following the separation of “is”es and “ought”s. Human judges and the hypothetical AI judges both would have flaws considering what we consider what “ought” to be. We simply suggest, the line of questioning brought by Legal Realism is suitable to study the decisions that would be made by AI models, as separate from “ought”s, following the Realist spirit. We have noted the so-called success of these models in their predictions simply to emphasise the need for a theoretical framework to understand them, as their accuracy implies incentives for their real-world applications in the future.

In the following section we will discuss the compatibility of the legal decisions that would be made by these models with the Legal Realist outlook. To that end we will discuss (1) the lack of AI models in a legal decision making setting, and (2) the compatibility of Legal Realist views with the application of AI models.

³¹⁹ Hilal Albal Ulaş, “Yargı Kararlarının Gerekçeli Olması Üzerine Bir Değerlendirme”, *Hacettepe Hukuk Fakültesi Dergisi*, Vol. 11, No. 2, 2021, p. 1245. DOI: 10.32957/hacettepehdf.958575. (Access Date: 29.03.2024)

A. WHERE AI MODELS LACK

According to Realist writers, judges act on their hunches and move backwards to generate a rational explanation for their judgements. This, however, only encompasses the delivery of the verdict. The role of the judge is beyond this in a standard court setting. Judge is often also responsible for things such as retrieval of evidence. What we mean by legal decision making, on the other hand, is simply the assigning of a verdict. These models cannot perform any other duties of a judge either. They are simply emulators of human judges in reaching verdicts. Thus, their use in a legal decision making setting will be limited to that, as our discussion is.

When considering how humans reach decisions, there are steps that human judges take, as various views would suggest. The models do not follow any steps towards the decision such as determining the facts, deciding on the rules which may apply, interpreting those rules or their application.³²⁰ As we have discussed, AI models are not designed to understand any facts or rules, they cannot understand or extract context. They simply take all the information available and predict what a human judge would do.

Human judges have other inclinations in reaching decisions, such as establishing justice. This view is shared by Realists as well. We have discussed in the section II.B. how the purpose of justice affects judges, not just as an ought but also as an is. AI models have no underlying understanding of concepts such as justice or fairness. What they can at most do is reproduce the tendencies of human judges without a deeper understanding of what they did and why they did it. It has also been discussed that being a judge requires

³²⁰ Pound, p. 945.

abilities such as research, logic, syllogism, analytical abilities, and understanding, and it is uncertain if AI judges would ever have these abilities.³²¹

There also has been discussions surrounding the advantages and disadvantages of AI systems in legal use. **Hikmet Bilgin** notes that it is not possible to guarantee the transparency and accountability of the AI since an appeal of an AI case to another AI appellate court is not suitable for changing the decision.³²² **Bilgin** argues so long as there are no clear answers regarding whether the justice has been delivered, AI would not be able to meet the expectations regarding public benefit expected of it.³²³ **Bilgin** also notes possible advantages of AI as follows:

In this way, all jurisprudence on a particular legal issue can be evaluated within seconds through systems based on artificial intelligence. Thus, one of the most important guarantees offered by the trial supported by artificial intelligence is that similar decisions will be given in similar cases on similar subjects, compatible with each other. In addition, it is to prevent individuals in the same situation from receiving different sentences by different judges due to subjective and personal effects, or from contradictory judgments that may arise because one of two people in the same situation wins the case while the other loses. In this way, the loss of time and effort that will occur due to conflicting provisions being corrected by higher-level judicial authorities or reversed and sent back will be minimized, and legal security and respect for the judiciary can be maintained.³²⁴

³²¹ Oğuz Gökhan Yılmaz, “Yargı Uygulamasında Yapay Zeka Kullanımı –Yapay Zekâ Hâkim Cübbesini Giyebilecek Mi?”, *Adalet Dergisi*, 66. Sayı, 2021/1, p. 402.

³²² Hikmet Bilgin, “Yapay Zekânın Mahkeme Kararlarında Kullanımına Uluslararası Bir Bakış ve Robot Hâkimler Hakkında Düşünceler?”, *İnönü Üniversitesi Hukuk Fakültesi Dergisi*, Cilt: 13, Sayı: 2, 2022, p. 413.

³²³ Bilgin, p. 413.

³²⁴ Bilgin, p. 412.

Although, through this advantages and disadvantages, **Yıldırım et al.** warns that “since replacing legal decision making with algorithmic models may have extremely harmful consequences, their most appropriate use for the moment may be to serve as decision support systems that will help speed up judicial processes.”³²⁵ **Andrew C. Micheals** also notes on this issue: “Replacing the human judiciary with an automated one that instantaneously generates an answer would erase the period of suspended conclusion during which societal legal thought takes place.”³²⁶ 62

The first point of criticism towards the use of AI is the one voiced by Scherer: how AI models create a judiciary setting that is dominated by previous decisions. This adherence to precedent is criticised by **Frank** in human judges as well.³²⁷ This is one of the most problematic aspects of usage of AI models in legal decision making. Since AI models are trained on previous decisions, they reiterate the previous mistakes made by judges, and follow the systemic biases that are in their data.

The second point of criticism would be the focus on words by AI. This issue is again criticised by **Frank** by calling it a word and rule fetishism.³²⁸ Decisions made by AI are focused on the words, rules, and precedents. Even though the summary of facts of a case is in the data, AI does not create new law as **Frank** calls for,³²⁹ and only interacts with the facts of a case through the words in the summary of those facts. With that, AI also fails to recognise the specific facts of a case, and to take those into consideration.

Another point where AI lacks as compared to the human judges is the knowledge a judge has accumulated. A judge is expected to (and does) use legal, ethical, and technical

³²⁵ Yıldırım et al. p. 484.

³²⁶ Andrew C. Micheals, “Artificial Intelligence, Legal Change, and Separation of Powers”, *University of Cincinnati Law Review*, Vol. 83, 2020, p. 1095.

³²⁷ Gürkan, p. 58.

³²⁸ Gürkan, p. 58.

³²⁹ Gürkan, p. 58.

knowledge. Even if we are to deny the human judge all faculties of reason, the hunches of a judge would be shaped by their previous knowledge and experience. This is discussed in part by **Radin**, in asking the questions of whether the judge's knowledge of the parties beforehand would influence the decision.³³⁰ In some sense, this can also be characterised as the "character" of the judge. On the other hand, the AI only has knowledge as far as the case data it has been trained on. As such, it will lack these faculties that would be utilised in giving a just verdict.

Through the "virtue" of these lacks, AI models used in legal decision making will only be able to reestablish previous law, in a closed and conservative setting. This can be made a point of criticism from a variety of standpoints. Even in non-legal applications the biases and prejudices embedded in AI have been problematic and made issue in the literature. Legal decisions by their nature are far more impactful on human lives. With that, in the case that these models are used for legal decision making, the biases that are currently in our judicial decisions will be engraved and reestablished.

B. APPLYING REALISM TO AI

We have established that our understanding of Legal Realism dealt with law in the sense of study of it. Then, we moved on to establishing how Legal Realism approached legal decision making and noting the points which would help us see why application of Legal Realist questioning would be compatible with legal decision making with AI. In the following section, we will discuss how AI compares in legal decision making to the extent of legal decision making as it was understood by Legal Realists and then we will see how the Legal Realist line of study would be applied to the usage of AI models in question.

³³⁰ Radin, p. 357.

Realist view is that legal decision making is not simple application of rules to facts, or as **Frank** puts it, $R \times F = D$.³³¹ **Frank** argues that the personality of the judge and the stimuli that works within that personality are far more important than the rules and the facts.³³² Even **Pound** who argued that the judge does apply rules to facts emphasised the role of hunches, as did many Realists as we have discussed in detail above.³³³ Realists also have noted that the judge might be moved by ideas such as justice or other political views.³³⁴

In understanding how AI makes legal decisions, we need to first note that the current AI models do not give decisions, they predict decisions. However, in their ability to predict what a human judge would do in a given case, their prediction can be used as a decision for that case, almost as if a self-fulfilling prophecy. Still, their method of prediction can be examined as if it were a method of decision making.

We have noted that the AI models work by clinging to words, processing them in such a manner that would enable it to predict a decision. These words do not come only from the decision but also from rules, or other details as it relates to the cases. With this, we see the questions of extra-legality in legal decision making with AI.

Most discussions we have mentioned focused on the sources of AI from a similar perspective, they have questioned the legal aspect of the data used. For example, **Aletras et al.** argued that since the law part of their data correlated less with the results, it could be so that Legal Realists were right in their arguments, although with caveats.³³⁵ Does that really imply that legality was not a part in predicting cases?

³³¹ Frank, “What Courts Do in Fact – Part One”, 1931-1932, p. 649.

³³² Frank, “What Courts Do in Fact – Part One”, 1931-1932, p. 662.

³³³ See Section II.B. for detailed discussions.

³³⁴ Gürkan, p. 24.

³³⁵ Aletras et al., p. 11.

There has been works that included legal text in their data and had success with their predictions, (e.g. **Long et al.** and **Kowsrihawwat et al.**). Still, **Mumcuoğlu et al.** employing similar methods albeit to another dataset, obtained similar results. Are we to accept importing data from legal texts has no impact, and does that mean no legal faculties are involved with the process?

We first need to note that legal decision making as it relates to Realism is not merely a question of extra-legality. Indeed, Legal Realism notes the usage of extra-legal variables in legal decision making, and so does AI utilise those factors in prediction. However, we must also note that AI, as we have discussed in the previous section, differs from the Legal Realist understanding of legal decision making in other areas. Still, this usage of extra-legal factors is similar in both Legal Realist understanding and AI models.

The other focus made by Legal Realists is the usage of hunches in legal decision making. AI models do not have hunches in the traditional sense. On the other hand, the function of hunches as described by Realists, as a translator of experience does exist in the AI models. As AI models are trained on previous decisions, it can be argued that this training data emulates the experience of a learned judge or lawyer and enables it to hunch out a decision.

Despite these *seeming* similarities, we cannot use the Legal Realist understanding of how judges made decisions and apply it to how AI predicts decisions. We neither know how judges make decisions in detail nor do we have a precise understanding of how AI makes predictions through statistics. As we have mentioned before, what Realism stands for as it relates to our study is the method of study of law. AI models cannot be compared to human judges, as AI models lack many features we see in human judges as we have discussed in the previous section. What can be done is that the decisions or predictions made by AI models can be examined by legal theory.

We have previously established points of interest for Realism. The first point we made was that law was uncertain. This approach is extremely convenient for legal theory as it relates to examining legal decisions made by AI. In accepting law is uncertain, Legal Realist theory looks to remedy it or to incorporate that uncertainty in theory. It so happens that AI models also work in uncertainty. We have noted that AI models work using probabilities, and thus we believe the Realist perspective is well equipped to deal with this probabilistic reality.

AI models do not use discretion, or any other traditional means that have been theorised before but relies on experience gathered in the form of training it—perhaps its character, in a way—in legal prediction or decision. The new case presented to it stimulates it through whatever method chosen, presents it with facts and laws, and it predicts an outcome. This way of deciding cases is akin to what Realists had thought was happening and what they were questioning and criticising.

Our second point was that Realists were concerned mainly with law in effect, with how lawyers could predict cases and how judges decided them. This area of interest proves to be useful in dealing with AI models as these models are purely practical tools. This area of interest thus coincides with what is happening with the new research. If AI models are used in legal decisions, any theory that would study how they decide cases must be focused on law in effect, as these models would be just that and nothing more.

The Realist agenda that sought to understand what officials were doing in administering law is hand to hand with any technology application in law. If these models were used not in a courtroom setting but for handling administrative requests, it still would fall to Legal Realism to examine this issue as the Realist theory seeks to understand the impact of legal application to people. In this, **Lon L. Fuller**'s conceptualisation of Realist understanding of law is then repeated: "a rule of law is a generalization about the

way judges act; or, as it has been phrased, the law consists in patterns of judicial behaviour”.³³⁶

We have abstained from defining law, and we have provided how leading Realists such as **Frank** and **Llewellyn** have done so as well. Still, this misrepresentation is not far off from our approach, Realists are interested in the way a judge acts and in patterns of judicial behaviour. AI models do interpret judicial behaviours made by judges, and act like Realists in that manner, but our issue is not with that. Our issue lies with the need to understand judicial behaviour, if it ever comes to that, by AI models, which is the interest of Legal Realism.

Our third point has been the removal of the rules from the focal point of legal study. Legal Realism provides a framework that enables for both legal factors and non-legal factors to be considered. We have to remember that **Holmes** put forward as to the means of study for lawyers as “body of reports, of treatises, and of statutes”.³³⁷ **Frank** also noted that rules might be among the factors that stimulate the judge and as such be point of inquiry.³³⁸ Realism simply asks for the consideration that maybe “rules of substantive law” should not be in the focal point of legal discussion.³³⁹

Studies which employed decision trees have used various metadata, in the case of **Virtucio et al.** it included some labels within the scope of traditional legal approaches, while the case of **Ruger et al.** arguably involved no legal basis as we have established previously. In any case, all models included extra-legal data, even if they have also included legal data. We also have previously mentioned how **Aletras et al.** mentions that

³³⁶ Fuller, p. 53

³³⁷ Holmes, p. 457.

³³⁸ Frank, “What Courts Do in Fact – Part One”, 1931-1932, p. 662.

³³⁹ Llewellyn, 1930, p. 442.

the correlation of “law” subsection of the data with the results was less significant than that of the “circumstances” subsection, which they say coheres with Realism.³⁴⁰

It is imperative to realise that this coherence does not imply the correctness of our thesis, it merely implies that judges *might* have been using extra-legal factors in their judgements and Realists *might* have been correct in their assumptions about how judges decided cases. Our argument is that Realism would be suited to analyse how legal decisions would be made by AI models. AI models openly employ extra-legal factors. Realist realisation that these extra-legal factors should also be content of legal theory is what our concern is. This approach in study, we believe, also coincides with how AI models work.

Our fourth point was the is and ought separation. **Llewellyn** explicitly states that there is no normative aspect to the Realist program.³⁴¹ Our opinion is that this lack functions in both supports and detriments Realism as a theory to be applied to usage of AI models. As AI is a new technology, there is a need for ethical study surrounding its applications. What *ought* to be done by this technology is important. As such, Realism seems to lack in addressing this issue. We have discussed where AI models lack and how their usage might contribute to injustice and reinforcement of biases.

On the other hand, the is ought separation inherent to the Realist study also is useful in understanding AI models. Since Realists focus on what is and not what ought to be, a Realist study would paint an objective picture of legal decision making by AI without any prejudice for further ethical research to be made. **Frank**’s comment that “if someone comments that someday Ramsay MacDonald will die, the truth of his statement is quite apart from the question whether he is glad, sorry, or neutral with respect to Mr.

³⁴⁰ Aletras et al., p. 12.

³⁴¹ Llewellyn, 1931, p. 1254.

MacDonald's unavoidable demise"³⁴² is parallel to our approach. AI models may be lacking, yet if they are used, we need to understand how they work.

Our last point was the employment of psychology in understanding judges. We have repeatedly mentioned how we consider Realism as it would apply to AI models, not as it applied to human judges. The translation of this approach would be adherence to computer science. If the discipline of law aims to understand how decisions are made in a scenario where AI models make decisions, interdisciplinary research is a must.

For our purposes, we think Realism is compatible for analysis of legal decision making with AI models. It is compatible in the questions it asks to the subject matter, it is compatible in its chosen subject matter, and it is compatible in its outlook towards the matters which it deals with. Questions of Legal Realism will help to explain machinations of AI models, Legal Realism is interested in legal decision makers and their behaviours, and Legal Realism accepts the uncertainty in law.

In accepting Legal Realism as a proper theory for understanding legal decision making with AI, we also prophesize that Legal Realism will be reborn if these tools come to be used for that purpose. We cannot know for certain if these tools will ever be used as they are in that setting; yet we predict that they will be. To quote **Radin**, albeit in a different context: "Our business is prophecy, and if prophecy were certain, there would not be much credit in prophesying."³⁴³

³⁴² Frank, "What Courts Do in Fact – Part Two", 1931-1932, p. 774.

³⁴³ Radin, p. 362.

CONCLUSION

Leiter voices his wish for a reborn Realism in saying “As we approach the turn of the century, I hope we may see Realism reborn yet again, this time as a naturalized approach to jurisprudence.”³⁴⁴ We believe that Realism may be reborn through the need to understand the new landscape jurisprudence will find itself in through the changes in the world in the form of technological advancements. AI applications find increased use in many cases and in legal decision prediction the new technologies have managed varying degrees of accuracy in their mimicry, with the best models managing around 90% accuracy, and others averaging around 70%. **Poppe** too voices the same fortune: “I foresee the increasing role of technology in law.”³⁴⁵

In **Mumcuoğlu et al.**, the corpora shows that more than one third of cases to civil court of appeal were admitted,³⁴⁶ *implying* a similar accuracy rate from local courts, who, *in some sense* try to imitate or predict the decisions of the higher courts in their decision making, which is not to say all these decisions should assumed to be “correct” or “accurate” in the traditional legal sense either. We believe these results might mean that technology might take over, in some future, the decision making process altogether.

In order to make sense of the new world as it is, we need a theory that studies the world as it is. We have discovered how Realism does this, and how Realism approached the questions of legal decision making. Judges, as they are human, provide no ease of study in understanding how they do what they do in their immense complexity. A new legal decision making, conducted by AI emulating these complex machines needs

³⁴⁴ Leiter, 1997, p. 315.

³⁴⁵ Poppe, p. 192.

³⁴⁶ Mumcuoğlu et al., p. 8.

thorough questioning due to the combination of our lack of understanding regarding both AI and humans in legal decision making.

We made the comparison of AI models to Pythia in the sense that neither consulted logic, at least logic as we are used it. We have examined three types of AI models and explained how they worked on a conceptual level and saw that all of them worked in some sense to recognise patterns in data as to be able to extract those patterns and use them in probabilistic calculations in determining what class an object belongs to or what the next probable token is. In this sense, we believe they are reminiscent of humans, as noted by some as a “similarity” between the Realist study and AI applications. **Mattson**, in an article titled “Superior pattern processing is the essence of the evolved human brain” published in 2014, argues what the title suggests: “the human brain is particularly advanced in its ability to fabricate new patterns”.³⁴⁷ It should be no surprise that the human brain resembles to the machines, as our own intelligence should be one of the only places that we could draw inspiration from in the quest for “artificial” intelligence.

Frank’s expressed grief as to the unlikelihood of success for a Movement for the Abolition of Ambiguous Words³⁴⁸ echoes the motivations of this study. However, it is almost too easy to argue, (maybe with ill intent) that *all* words are ambiguous. If we are to apply a pragmatistic approach to Law, we must also consider the implications of applying the same approach to all else. In the case of words, they are “reduced” to but vessels of meaning.

³⁴⁷ Mark P. Mattson, Superior Pattern Processing is the Essence of the Evolved Human Brain, *Frontiers in Neuroscience*, Vol. 8, 2014, p. 13.

³⁴⁸ Frank, “What Courts Do in Fact – Part One”, 1931-1932, p. 645.

The human condition is that we have to live in communities in order to survive better, and a community needs communication. Words are extremely useful tools in communication which can relay meaning from one individual to another. Law is just one extent of communication, and an area of use for words.

Let us imagine a scenario, where we need no words, a world in which we can communicate telepathically and relay any meaning to one another in an instant with utmost certainty. It is difficult to imagine such a scenario, not only because of the unlikelihood of ever achieving such precise communication, but primarily because we do not usually have a complete understanding of what we wish to mean at any given instance. We do not always have a complete understanding of what we wish to mean because we do not have a complete model of the world around us, nor ourselves.

The argument here is not one of scepticism, it is that ambiguity of words or ambiguity of communication in general is useful to us so that we can communicate without the need for understanding all things. On the other side, a society comprised of Laplace's Demons would have no use for words, still, they would have no use for living either, as they would know the past, the present, and the future as it is; and temporality would lose its meaning for them. So, it seems **Frank's** grief is uncalled-for, a Movement for the Abolition of Ambiguous Words would call for an abolition of our human experience.

The problem of relaying meaning through words is twofold, not only we cannot always express what we wish to express exactly, the receiver of our message also often fails in interpretation of the message, so the meaning gets doubly corrupted. Adding to that the fact that we had an incomplete understanding of the world, it is almost impossible to recover the meaning to certainty. Conceiving rules, writing them, and then

reinterpreting them, reiterating them in decisions and predictions is almost a game of telephone played by lawmakers, judges, jurists, and lawyers in an ever-desperate fashion.

Realist outlook we have (un)covered is *realist* towards this reality; Realism manages to embrace this ambiguity instead of rejecting it, as **Frank** did in calling for avoiding the arguments regarding the meaning of Law.³⁴⁹ Realist outlook, in general, called for a scepticism towards the certainty of rules. **Llewellyn's** parody stating “Before rules, were facts; in the beginning was not a Word, but a Doing”³⁵⁰ is what we have argued in the beginning of the introduction of this thesis: words only describe doings. Now, we add to that: words only describe doings, and they do it poorly.

Legal Realism is not a remedy for this problem of ambiguity, it simply points it out so that we can be aware of it. In being aware of it we can have a more accurate model of the world, a better framework for evaluating possible remedies, and realise when it is the cause of any other issues. Legal Realism is a set of warnings which try to prevent those who deal with Law from misunderstanding it, which is why we believe it should be applied to AI models used in a legal decision making setting.

Realism has one simple wish: to understand legal affairs as they are apparent in the world around us, without seeking to create a periphery of what it entails. **Fuller's** critique in *Law in Quest of Itself* makes it clear that there is a need for some sense of limit to legal examination.³⁵¹ We would argue that, that limit is whatever is considered to be related to legal matters. To that end, Legal Realism seems to approach Law as a scientific endeavour which creates knowledge as to make a prediction given the same circumstances, limited by a vague domain and lacking a clear methodology as to the how.

³⁴⁹ Frank, “What Courts Do in Fact – Part One”, 1931-1932, p. 645.

³⁵⁰ Llewellyn, 1931, p. 1222.

³⁵¹ Fuller, p. 52-63.

To the extent that we seek to understand law, we need to be prepared towards change in law. **Llewellyn** talked of a “conception of law in flux”³⁵², while he did not mean necessarily how application of law might change; this idea of change required preparedness. If such a major change as non-human decision making comes to be, we need tools to understand it, regardless of its success or what we think ought to happen. Throughout this paper, we tried to embrace the separation of is and ought, and we do not suggest at any point that legal decision making should be left to AI models, what we have argued that if this happens Realism is the best tool we have to understand it; and as such we would witness its renaissance.

In concluding our work, we also feel the wish to note a question beyond our study, that is if law is a science. It is for sure not for us to answer that question. However, we can propose an idea. Law can include scientific, hermeneutic, and normative aspects. As we have mentioned before, Realists also argued about the various meanings that the word law conveys. In doing this, we position Legal Realism in a place that deals with the practical aspect of Law which emphasises that Law does allow such a perspective. We see that there are other subjects of Law that have been made subject of empirical study other than adjudication, and this is also compatible with this view. The hermeneutic aspect of Law includes the interpretation of legal texts, almost as part of any literature. Normative aspect of Law, on the other hand, with much critique coming from **Fuller**, would deal separately with its oughts.

³⁵² Llewellyn, 1931, p. 1236.

In chess, the number of possible moves is estimated to be more numerous than the number of particles in the universe.³⁵³³⁵⁴ It is still possible, however, for us to play chess. It is impossible to consider all possible advantages to the end, and make a decision based on pure calculation. Instead, advantageous situations are considered through heuristic approaches, by which we mean application of past experience of trials and errors to the situation at hand. The complexity of legal tasks demand of a lawyer or a judge, implies that neither can apply pure calculation to a case. If that were possible, that would also mean that task would be simple enough that we would have “calculators” instead of experts who are trained on legal matters for years.

The state of the world is a state of causes and effects. Each small action contributes to other actions and a new state is thus formed in each consecutive moment. **Pierre-Simon Laplace**, in his *A Philosophical Essay on Probabilities* talks of an invented being which would be able to deduce from one instance of the universe a previous one or the next:

We ought then to regard the present state of the universe as the effect of its anterior state and as the cause of the one which is to follow. Given for one instant an intelligence which could comprehend all the forces by which nature is animated and the respective situation of the beings who compose it—an intelligence sufficiently vast to submit these data to analysis—it would embrace in the same formula the movements of the greatest bodies of the universe and those of the

³⁵³ Louis Victor Allis, *Searching for Solutions in Games and Artificial Intelligence*, Wageningen: Ponsen & Looijen, 23 Sep. 1994, p. 171.

³⁵⁴ Melvin M. Vopson, “Estimation of the Information Contained in the Visible Matter of the Universe”, *AIP Advances* 11, 105317, 2021 p. 5.

lightest atom; for it, nothing would be uncertain and the future, as the past, would be present to its eyes.³⁵⁵³⁵⁶

A predictor must aspire to be such a demon; to gather as much data as possible, for the lack of ability to gather all of it, and to guess what will follow at a certain point in the future. As such, this is our Realist inspiration.

The legal decision prediction algorithms by AI not only promise better access to justice through faster adjudication in case they are used as legal decision making machines, but they also could be used as control mechanisms for incoherent legal decisions by the courts. Many authors have argued that AI models should be used as helpers for the courts instead of “robot judges”. While there are problems of bias in AI systems, these biases are only reflective of the data they are trained on, the bias of the previous decisions. As a technical problem, if this can be dealt with, we would have less bias in the future. Humans have their reasons to have bias, machines do not. We do not wish to argue normatively, lest **Hume**’s guillotine condemns us. Still, *if* we are to ask for a better understanding of the legal affairs, or even better access to “justice” —so much as it can be delivered by the courts— we must chant **Nazım Hikmet**’s chant: “I want to be mechanised”.³⁵⁷

³⁵⁵ Pierre Simon, Marquis de Laplace, *A Philosophical Essay on Probabilities*, tr. Frederick Wilson Truscott and Frederick Lincoln Emory, First ed., New York: John Wiley & Sons; London: Chapman & Hall, Limited, 1902, p. 4.

³⁵⁶ The original text is as follows: “Nous devons donc envisager l’état présent de l’univers comme l’effet de son état antérieur et comme la cause de celui qui va suivre. Une intelligence qui, pour un instant donné, connaîtrait toutes les forces dont la nature est animée et la situation respective des êtres qui la composent, si d’ailleurs elle était assez vaste pour soumettre ces données à l’Analyse, embrasserait dans le même formule les mouvements des plus grands corps de l’univers et ceux du plus léger atome : rien ne serait incertain pour elle, et l’avenir, comme le passé, serait présent à ses yeux.” Pierre Simon, Marquis de Laplace, *Théorie Analytique des Probabilités*, Paris, 1820, p. VI-VII.

³⁵⁷ Nazım Hikmet Ran, “Makinalaşmak”, *Nazım Hikmet Bütün Şiirleri*, Ed. Güven Turan, Fahri Güllüoğlu, İstanbul: Yapı Kredi Yayınları, 2008, s. 38.

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ABSTRACT

Legal decision making with AI is becoming more and more probable as research progresses. It includes mechanisms for prediction that are not compatible to be understood through traditional legal theories as they are being developed right now. Legal Realism was a critical approach to traditional legal theories which have managed to propose alternative ways to understand how legal affairs were conducted with an emphasis on prediction of legal decisions as the focal point of legal study, instead of rules.

The first chapter of the thesis discusses the fundamental ideas of legal realism as they relate to the debate. The second chapter examines research on legal decision making with AI and how the current models work at the surface level. Last chapter argues and concludes that Legal Realism provides an adequate framework for legal decision making with AI from a legal study perspective.

ÖZET

Yapay zekâ ile hukuki karar verme, arařtırmalar ilerledikçe giderek daha olası hale geliyor. řu anda geliřtirilmekte olan aralar geleneksel hukuk teorileri ile anlařılması mümkün olmayan tahmin mekanizmalarını içermektedir. Hukuki Realizm, kurallar yerine hukuk alıřmasının odak noktası olarak hukuki kararların öngörölmesine vurgu yaparak hukuki işlerin nasıl yürütüldüğünü anlamak için alternatif yollar önermeyi başaran geleneksel hukuk teorilerine eleřtirel bir yaklařımdı.

Tezin ilk bölümünde Hukuki Realizmin temel fikirleri konuyla baėlantılı olarak tartıřılmaktadır. İkinci bölümde, yapay zekâ ile hukuki karar alma konusundaki arařtırmalar ve mevcut modellerin yüzeysel düzeyde nasıl alıřtığı inceleniyor. Son bölümde hukuki gerekiliğın hukuki alıřma perspektifinden yapay zekâ ile hukuki karar alma için yeterli bir ereve saėlayıp saėlamadığı tartıřılıyor ve Hukuki Realizmin bu durum için uygun bir ereve saėladığı sonucuna varılıyor.